

Oracle Linux Virtualization Manager

Exam Practice Test Collection

Exam 1Z0-1170

Oracle Linux Virtualization Manager: Administration

11 Practice Tests | 515 Total Questions

Covers All 11 Exam Objectives

Questions with Answer Keys

This document is intended as a study aid for Oracle Partners preparing for the 1Z0-1170 certification exam.

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How to Use This Document

Each section in this document maps to a training session from the Oracle course:

<https://mylearn.oracle.com/ou/course/oracle-linux-virtualization-manager-administration/158560>

The course includes 11 training sessions. Watch one session, then take the matching practice test before moving on:

1. Architecture and Overview
2. Installing the Engine
3. Understanding Core Components
4. KVM Host Installation
5. Managing Storage Domains
6. Network Management
7. User Management
8. Optimization Practices
9. Administrrating Virtual Machines
10. Managing Events and Logs
11. OLVM Recovery Management

Study Workflow

1. Watch one training session from the Oracle course.
2. Open the matching section in this document and work through the questions without looking at the answer key.
3. The answer key is at the end of each section — scroll past the questions first.
4. Score yourself using the interpretation guide provided.
5. Review the questions you got wrong and revisit those topics in the training.
6. Retake the section if you scored below 80%.
7. Move on to the next training session and repeat.

Section 1: Architecture & Overview

60 Questions

FUNDAMENTALS & CORE CONCEPTS

1. What is the primary database schema name used by the oVirt engine in Oracle Linux Virtualization Manager?

- A. ovirt_db
- B. engine
- C. ovirt_engine
- D. manager_db

2. Which two Intel hardware extensions does Oracle Linux Virtualization Manager support for enhanced performance? (Choose 2)

- A. Intel VT-x
- B. Intel SGX
- C. Intel VT-d
- D. Intel AES-NI

3. What is the function of Intel VT-x in virtualization?

- A. It handles input/output device virtualization
- B. It creates a virtual processor for each virtual machine
- C. It manages network traffic between VMs
- D. It provides encryption for virtual disks

4. What is the purpose of Intel VT-d in Oracle Linux Virtualization Manager?

- A. It provides processor-level virtualization
- B. It allows VMs to directly use devices like graphic cards and network adapters
- C. It manages memory allocation between VMs
- D. It handles virtual machine migration

5. What is "CPU pinning" (hard partitioning) used for in Oracle Linux KVM?

- A. Increasing CPU performance for all VMs
- B. Assigning specific vCPUs to certain physical CPU threads or cores
- C. Automatically balancing CPU load across hosts
- D. Reducing power consumption of physical CPUs

6. What is the primary licensing benefit of using CPU pinning for Oracle products?

-
- A. It reduces the total cost of Oracle licenses
 - B. It ensures licensing is based only on the number of physical cores in use
 - C. It eliminates the need for Oracle licenses
 - D. It allows unlimited virtual machines per core

ARCHITECTURE COMPONENTS

7. What is the core component that serves as the backbone of Oracle Linux Virtualization Manager?

- A. KVM hypervisor
- B. VDSM agent
- C. oVirt engine
- D. PostgreSQL database

8. Which three tasks does the oVirt engine perform? (Choose 3)

- A. Discovering KVM hosts
- B. Running virtual machine processes
- C. Configuring storage for virtualized data centers
- D. Configuring networking for virtualized data centers
- E. Providing hardware emulation for VMs

9. Oracle Linux Virtualization Manager is built from which open-source project?

- A. OpenStack
- B. oVirt
- C. Proxmox
- D. XenServer

10. What are the two main interfaces provided by Oracle Linux Virtualization Manager for management? (Choose 2)

- A. Command-line interface (CLI)
- B. Web-based UI
- C. REST API
- D. SNMP console

11. What happens to a virtual machine if the oVirt engine goes offline?

- A. The VM automatically suspends
- B. The VM continues to run on the KVM host
- C. The VM is migrated to another host
- D. The VM shuts down gracefully

SELF-HOSTED ENGINE

12. What is a Self-Hosted Engine in Oracle Linux Virtualization Manager?

-
- A. An engine that requires a dedicated physical server
 - B. A management engine that runs as a VM directly on the virtualization hosts
 - C. An engine that runs in Oracle Cloud Infrastructure
 - D. A backup engine for disaster recovery

13. What is the primary benefit of using a Self-Hosted Engine deployment?

- A. It requires more hardware resources
- B. It eliminates the need for a separate management server
- C. It increases the complexity of deployment
- D. It requires Oracle Linux Enterprise Kernel

14. How does Self-Hosted Engine enhance resilience?

- A. By requiring dedicated backup servers
- B. By eliminating single points of failure with the engine distributed across multiple hosts
- C. By automatically creating VM backups <https://oracle.sharepoint.com/sites/s2s-mobile-phones/>
- D. By using RAID storage configurations

15. In a Self-Hosted Engine deployment, where is the Engine VM stored?

- A. On the local disk of the first host
- B. On shared storage accessible by all hosts
- C. In Oracle Cloud Infrastructure
- D. On a USB drive connected to the host

16. What component monitors and manages the Self-Hosted Engine VM on each host?

- A. oVirt engine service
- B. VDSM daemon
- C. HA agent (ovirt-ha-agent)
- D. PostgreSQL database

DATA WAREHOUSE & DATABASES

17. How many PostgreSQL databases are used by Oracle Linux Virtualization Manager?

- A. One
- B. Two
- C. Three
- D. Four

18. What is the name of the primary database created during engine configuration?

- A. ovirt_engine
- B. engine
- C. manager
- D. ovirt_db

19. What is the name of the second database used for historical data?

- A. engine_history
- B. ovirt_engine_history
- C. dwh_database
- D. historical_data

20. What must be installed to create the ovirt_engine_history database?

- A. ovirt-engine-dwh package
- B. postgresql-history package
- C. engine-history module
- D. ovirt-database-tools

21. What is the primary purpose of the Data Warehouse (DWH) component?

- A. To store virtual machine disk images
- B. To collect and store historical performance and usage data
- C. To manage the storage pool metadata
- D. To backup the engine database

MANAGEMENT & INTERFACES

22. What information does the Oracle Linux Virtualization Manager dashboard provide? (Choose 3)

- A. VM counts and status
- B. Host counts and status
- C. Cluster information
- D. Individual user login history
- E. Storage status and performance metrics

23. Which feature allows Oracle Linux Virtualization Manager to integrate with other management systems?

- A. Web-based UI
- B. SSH access
- C. REST API
- D. SNMP traps

24. What is the primary advantage of Oracle Linux Virtualization Manager being open-source?

- A. It requires no configuration
- B. It provides cost savings with no licensing fees
- C. It only runs on Oracle hardware
- D. It cannot be customized

25. What is the relationship between vCPUs and physical CPUs in Oracle Linux KVM?

- A. Each physical CPU can only support one vCPU

-
- B. The guest VM decides which tasks run on vCPUs, and the hypervisor decides which physical CPU cores the vCPUs use
 - C. vCPUs must always be equal in number to physical CPUs
 - D. Physical CPUs are not visible to the hypervisor

KVM HOST ARCHITECTURE COMPONENTS

26. What is the role of VDSM in Oracle Linux Virtualization Manager?

- A. It provides hardware emulation for virtual machines
- B. It acts as a host agent running continuously as a daemon on the KVM host
- C. It manages the PostgreSQL database
- D. It provides the web-based user interface

27. How does the oVirt engine communicate with KVM hosts?

- A. Through SSH connections
- B. Through the VDSM service
- C. Through direct kernel calls
- D. Through the PostgreSQL database

28. Which three tasks does VDSM perform on the KVM host? (Choose 3)

- A. Managing virtual machines
- B. Providing web browser access
- C. Managing networks
- D. Managing storage
- E. Running the oVirt engine

29. What is the function of libvirt in the KVM host architecture?

- A. It emulates hardware components for VMs
- B. It provides an API for managing various hypervisors including Oracle Linux KVM
- C. It stores virtual machine configurations
- D. It manages user authentication

30. What does QEMU (Quick Emulator) provide in the virtualization stack?

- A. Network connectivity between VMs
- B. Hardware component emulation such as CPU, memory, network, and disk devices
- C. User authentication services
- D. Database management

31. Where does KVM operate within the system?

- A. In user space as an application
- B. In the kernel space
- C. On a separate management server
- D. In the cloud

32. Where do virtual machines run as processes?

- A. In kernel space as KVM modules
- B. As QEMU processes in user space
- C. On the oVirt engine server
- D. In the VDSM daemon

33. What is the relationship between VDSM and libvirt?

- A. VDSM replaces libvirt completely
- B. VDSM relies on libvirt to manage the lifecycle of VMs and collect statistics
- C. libvirt runs inside VDSM
- D. They operate independently without interaction

ADMINISTRATION PORTALS

34. Which three portals are available in Oracle Linux Virtualization Manager? (Choose 3)

- A. Administration Portal
- B. Developer Portal
- C. VM Portal
- D. Monitoring Portal
- E. Database Portal

35. What is the primary purpose of the Administration Portal?

- A. To allow end users to access their VMs only
- B. To serve as a nerve center for administrators to oversee, create, and maintain every component of the virtual ecosystem
- C. To monitor network traffic exclusively
- D. To backup virtual machines

36. How is the Administration Portal accessed?

- A. Through a dedicated client application only
- B. Through any web browser
- C. Through SSH command line only
- D. Through the VM console

37. What is the primary purpose of the VM Portal?

- A. To provide full administrative control over all OLVM components
- B. To cater to users who primarily engage with virtual machines, offering a user-friendly interface for basic VM management
- C. To monitor system performance only
- D. To configure storage domains

38. Which operations can users perform in the VM Portal? (Choose 3)

- A. Creating, editing, and removing virtual machines

-
- B. Starting, stopping, and migrating virtual machines
 - C. Modifying host hardware configurations
 - D. Accessing detailed VM information
 - E. Installing the hypervisor kernel

39. Who determines the capabilities available to users within the VM Portal?

- A. The users themselves
- B. Oracle support team
- C. System administrators
- D. The oVirt engine automatically

40. Which two protocols can be used to connect to virtual machines from the VM Portal? (Choose 2)

- A. SSH
- B. VNC
- C. SPICE
- D. RDP (for Windows VMs only)

MONITORING & GRAFANA

41. What is the primary purpose of the Monitoring Portal in OLVM?

- A. To create new virtual machines
- B. To empower administrators with comprehensive monitoring capabilities for health and performance
- C. To manage user permissions
- D. To configure storage domains

42. Which open-source analytics platform does Oracle Linux Virtualization Manager integrate with for enhanced monitoring?

- A. Kibana
- B. Prometheus
- C. Grafana
- D. Nagios

43. Where does Grafana retrieve its monitoring data from in OLVM?

- A. Directly from the KVM hosts
- B. From the engine data warehouse
- C. From virtual machine logs
- D. From network switches

44. What can administrators create in Grafana for monitoring?

- A. Virtual machines
- B. Storage domains
- C. Customized dashboards tailored to specific monitoring needs

D. User accounts

45. Which metrics can be monitored through Grafana dashboards? (Choose 3)

- A. CPU utilization
- B. User login history
- C. Memory usage
- D. Storage capacity
- E. VM creation timestamps

COCKPIT WEB INTERFACE

46. What is Cockpit in Oracle Linux Virtualization Manager?

- A. A database management tool
- B. A web interface that empowers users to monitor KVM host resources and perform administrative tasks
- C. A virtual machine migration tool
- D. A backup solution

47. How is Cockpit made available in OLVM?

- A. It is automatically installed with the oVirt engine
- B. It must be installed and activated separately
- C. It comes pre-configured in all KVM hosts
- D. It is only available through Oracle support

48. Which two methods can be used to access the Cockpit web interface? (Choose 2)

- A. Through the Administration Portal
- B. Through email notifications
- C. By establishing a direct connection to the host
- D. Through the VM console
- E. Through SNMP traps

49. Which metrics can users monitor through the Cockpit web interface? (Choose 3)

- A. CPU usage of the KVM host
- B. Individual VM application logs
- C. Memory utilization of the KVM host
- D. Disk space on the KVM host
- E. Network activity of the KVM host

ADVANCED FEATURES

50. What is "Live Migration" in Oracle Linux Virtualization Manager?

- A. Moving VM disk images between storage while the VM is powered off
- B. Moving running VMs to other servers seamlessly without downtime

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- C. Backing up VMs to tape drives
 - D. Converting physical servers to virtual machines

51. What is "Storage Live Migration" used for?

- A. Moving VMs between data centers
- B. Moving virtual disks of running VMs between storage devices without disrupting operations
- C. Backing up storage arrays
- D. Encrypting storage volumes

52. What technology does OLVM integrate to patch the kernel and QEMU without downtime?

- A. Ansible
- B. Puppet
- C. Ksplice
- D. Chef

53. How does OLVM provide backup capabilities for virtual machines?

- A. Using tape backup systems
- B. Through cloud synchronization only
- C. Using snapshots that capture a consistent view of running VMs at a specific moment in time
- D. Through manual file copying

54. Which two disaster recovery solutions does Oracle Linux Virtualization Manager support? (Choose 2)

- A. Active-Active
- B. Active-Standby
- C. Active-Passive
- D. Passive-Passive

55. In an Active-Active disaster recovery configuration, how are the sites connected?

- A. Through manual failover procedures
- B. Through stretched clusters, ensuring continuous application availability
- C. Through scheduled data transfers
- D. They are not connected

56. What authentication systems can Oracle Linux Virtualization Manager integrate with? (Choose 2)

- A. Local user accounts only
- B. LDAP
- C. Active Directory
- D. OAuth only

57. What is the maximum number of logical CPUs that the Unbreakable Enterprise Kernel (UEK) can support?

-
- A. 256
 - B. 512
 - C. 1,024
 - D. 2,048

58. What is the maximum amount of memory that the Unbreakable Enterprise Kernel (UEK) can support?

- A. 16 terabytes
- B. 32 terabytes
- C. 64 terabytes
- D. 128 terabytes

59. Which Java application server is the oVirt engine based on?

- A. Tomcat
- B. WebLogic
- C. WildFly
- D. JBoss EAP 6

60. What is the primary benefit of Oracle Linux KVM templates?

- A. They provide better security
- B. They cut down on installation time and ongoing maintenance costs
- C. They increase VM performance
- D. They reduce storage requirements

Answer Key — Architecture & Overview

1. B - engine
2. A, C - Intel VT-x and Intel VT-d
3. B - It creates a virtual processor for each virtual machine
4. B - It allows VMs to directly use devices like graphic cards and network adapters
5. B - Assigning specific vCPUs to certain physical CPU threads or cores
6. B - It ensures licensing is based only on the number of physical cores in use
7. C - oVirt engine
8. A, C, D - Discovering KVM hosts, Configuring storage, Configuring networking
9. B - oVirt
10. B, C - Web-based UI and REST API
11. B - The VM continues to run on the KVM host
12. B - A management engine that runs as a VM directly on the virtualization hosts
13. B - It eliminates the need for a separate management server
14. B - By eliminating single points of failure with the engine distributed across multiple hosts
15. B - On shared storage accessible by all hosts

-
16. C - HA agent (ovirt-ha-agent)
 17. B - Two
 18. B - engine
 19. B - ovirt_engine_history
 20. A - ovirt-engine-dwh package
 21. B - To collect and store historical performance and usage data
 22. A, B, C, E - VM counts/status, Host counts/status, Cluster information, Storage status/performance
 23. C - REST API
 24. B - It provides cost savings with no licensing fees
 25. B - The guest VM decides which tasks run on vCPUs, and the hypervisor decides which physical CPU cores the vCPUs use
 26. B - It acts as a host agent running continuously as a daemon on the KVM host
 27. B - Through the VDSM service
 28. A, C, D - Managing virtual machines, Managing networks, Managing storage
 29. B - It provides an API for managing various hypervisors including Oracle Linux KVM
 30. B - Hardware component emulation such as CPU, memory, network, and disk devices
 31. B - In the kernel space
 32. B - As QEMU processes in user space
 33. B - VDSM relies on libvirt to manage the lifecycle of VMs and collect statistics
 34. A, C, D - Administration Portal, VM Portal, Monitoring Portal
 35. B - To serve as a nerve center for administrators to oversee, create, and maintain every component of the virtual ecosystem
 36. B - Through any web browser
 37. B - To cater to users who primarily engage with virtual machines, offering a user-friendly interface for basic VM management
 38. A, B, D - Creating/editing/removing VMs, Starting/stopping/migrating VMs, Accessing detailed VM information
 39. C - System administrators
 40. B, C - VNC, SPICE
 41. B - To empower administrators with comprehensive monitoring capabilities for health and performance
 42. C - Grafana
 43. B - From the engine data warehouse
 44. C - Customized dashboards tailored to specific monitoring needs
 45. A, C, D - CPU utilization, Memory usage, Storage capacity
 46. B - A web interface that empowers users to monitor KVM host resources and perform administrative tasks
 47. B - It must be installed and activated separately
 48. A, C - Through the Administration Portal, By establishing a direct connection to the host
 49. A, C, D, E - CPU usage, Memory utilization, Disk space, Network activity (all of KVM host)
 50. B - Moving running VMs to other servers seamlessly without downtime
 51. B - Moving virtual disks of running VMs between storage devices without disrupting operations
 52. C - Ksplice
 53. C - Using snapshots that capture a consistent view of running VMs at a specific moment in time
 54. A, C - Active-Active, Active-Passive
 55. B - Through stretched clusters, ensuring continuous application availability

-
- 56. B, C - LDAP, Active Directory
 - 57. D - 2,048
 - 58. C - 64 terabytes
 - 59. C - WildFly
 - 60. B - They cut down on installation time and ongoing maintenance costs

Score Interpretation

- 50-60 correct: Excellent - Ready for this module
- 40-30 correct: Good - Review missed topics
- 30-20 correct: Fair - Study weak areas
- Below 20: Needs review - Re-read the training transcript

Section 2: Engine Installation

40 Questions

ENGINE HOST PREREQUISITES

- 1. What is the minimum Oracle Linux version required for installing the OLVM Manager (Engine)?**
 - A. Oracle Linux 7.5
 - B. Oracle Linux 8.5
 - C. Oracle Linux 9.0
 - D. Oracle Linux 6.8
- 2. Which two CPU technologies must be supported for the Engine host processor? (Choose 2)**
 - A. Intel VT-x
 - B. Intel SGX
 - C. AMD-V
 - D. ARM TrustZone
- 3. What type of CPU architecture is required for the Engine host?**
 - A. 32-bit x86
 - B. 64-bit x86
 - C. ARM64
 - D. PowerPC
- 4. Which six repositories must be enabled on the Oracle Linux system for OLVM Engine? (Choose 6)**
 - A. BaseOS Latest
 - B. AppStream
 - C. KVM AppStream
 - D. oVirt 44
 - E. oVirt 44 Extras
 - F. Gluster AppStream
- 5. Which additional repository channel is required specifically for configuring VDSM on the host?**
 - A. UEKR6
 - B. UEKR7
 - C. KVM Utils
 - D. Gluster 9

DEPLOYMENT SIZES

6. What are the MINIMUM hardware requirements for a SMALL deployment Engine? (Choose 3)

- A. 2 core CPUs
- B. 4 GB RAM
- C. 25 GB local writable disk
- D. 8 GB RAM
- E. 50 GB local writable disk

7. What are the RECOMMENDED hardware requirements for a SMALL deployment? (Choose 3)

- A. 2 cores
- B. 4 cores
- C. 16 GB or greater RAM
- D. 32 GB RAM
- E. 50 GB or greater local writable disk

8. How many KVM hosts can a SMALL deployment support?

- A. 1 to 5 hosts
- B. 5 to 10 hosts
- C. 10 to 20 hosts
- D. 20 to 50 hosts

9. How many virtual machines can a SMALL deployment support?

- A. Up to 20 VMs
- B. Up to 50 VMs
- C. Up to 100 VMs
- D. Up to 200 VMs

10. What are the recommended hardware requirements for a MEDIUM deployment? (Choose 3)

- A. 4 core CPUs
- B. 8 core CPUs
- C. 32 GB or greater RAM
- D. 64 GB RAM
- E. 100 GB or greater local writable disk

11. How many KVM hosts can a MEDIUM deployment support?

- A. 1 to 5 hosts
- B. 5 to 50 hosts
- C. 50 to 100 hosts
- D. 100 to 200 hosts

12. How many virtual machines can a MEDIUM deployment support?

- A. 50 to 100 VMs

-
- B. 50 to 500 VMs
 - C. 500 to 1000 VMs
 - D. 1000 to 2000 VMs

13. What are the recommended hardware requirements for a LARGE deployment? (Choose 3)

- A. 8 cores
- B. 16 cores or greater
- C. 64 GB or greater RAM
- D. 128 GB RAM
- E. 200 GB or greater writable disk space

14. How many KVM hosts can a LARGE deployment support?

- A. 10 to 50 hosts
- B. 50 to 200 hosts
- C. 200 to 500 hosts
- D. 500 to 1000 hosts

15. How many virtual machines can a LARGE deployment support?

- A. 500 to 1000 VMs
- B. 500 to 2000 VMs
- C. 2000 to 5000 VMs
- D. 5000 to 10000 VMs

FIREWALL REQUIREMENTS

16. What is the default port number for SSH access to the Manager?

- A. 21
- B. 22
- C. 23
- D. 3389

17. Which two ports are used for web interface and REST API access? (Choose 2)

- A. 80 (TCP)
- B. 8080 (TCP)
- C. 443 (TCP)
- D. 8443 (TCP)

18. What is the default port number for PostgreSQL database communication?

- A. 3306
- B. 5432
- C. 5433
- D. 27017

19. Which two ports are used for VDSM communication between the Engine and hosts? (Choose 2)

- A. 54320
- B. 54321
- C. 54322
- D. 54323

20. What is the port range used for SPICE protocol remote desktop connections?

- A. 3389 to 3400
- B. 5800 to 5900
- C. 5900 to 6923
- D. 8000 to 9000

21. What port is used for iSCSI storage domain access?

- A. 111
- B. 2049
- C. 3260
- D. 32803

22. Which port is used by the portmapper service for NFS?

- A. 111 (TCP and UDP)
- B. 2049 (TCP and UDP)
- C. 32803 (TCP)
- D. 3260 (TCP)

PREREQUISITE MODULES & PACKAGES

23. Which two appstream modules must be enabled before installing the OLVM Engine? (Choose 2)

- A. php:7.4
- B. pki-deps
- C. postgresql:13
- D. nodejs:14

24. What does PKI stand for in the context of OLVM?

- A. Package Key Infrastructure
- B. Public Key Infrastructure
- C. Private Key Interface
- D. Portal Key Integration

25. Which virt module should be DISABLED before installing OLVM Engine?

- A. virt:rhel
- B. virt:ol

- C. virt:kvm
- D. virt:8.5

26. Which virt module version should be ENABLED for OLVM Engine installation?

- A. virt:rhel
- B. virt:kvm_utils1
- C. virt:kvm_utils2
- D. virt:ol8

27. Which repository should be enabled for the latest base operating system packages?

- A. ol7_baseos_latest
- B. ol8_baseos_latest
- C. ol8_appstream
- D. ol8_UEKR7

28. What command is used to install the Oracle oVirt release package for Enterprise Linux 8?

- A. dnf install ovirt-release-el8
- B. dnf install oracle-ovirt-el8
- C. dnf install oracle-ovirt-release-el8
- D. dnf install ovirt-engine-release

ENGINE INSTALLATION & CONFIGURATION

29. What command is used to install the OLVM Engine package?

- A. yum install olvm-engine
- B. dnf install ovirt-engine
- C. dnf install olvm-manager
- D. yum install oracle-engine

30. What command is used to configure the OLVM Engine after package installation?

- A. ovirt-setup
- B. engine-setup
- C. olvm-configure
- D. manager-setup

31. What type of interface does the engine-setup command provide?

- A. Graphical user interface
- B. Web-based interface
- C. Interactive command-line with questions
- D. Silent installation with config file

32. What does the engine-setup command display after all questions are answered?

- A. Error log

-
- B. Summary of entered values
 - C. Installation progress bar
 - D. Database schema

33. What information is displayed after the engine-setup configuration is complete?

- A. System resource usage
- B. Details about how to log in to the Administration Portal
- C. List of available VMs
- D. Network topology diagram

ENGINE CONFIGURATION OPTIONS

34. Which five configuration groupings are part of the engine-setup process? (Choose 5)

- A. Database configuration
- B. Hardware configuration
- C. Network configuration
- D. Administration user setup
- E. Certificates and security
- F. Service configurations

35. What can the database configuration in engine-setup allow you to configure? (Choose 2)

- A. A local database instance
- B. A remote database
- C. MySQL database
- D. Oracle Database

36. What is the default administrative username created during engine-setup?

- A. root
- B. administrator
- C. admin
- D. ovirt

37. What is the name of the domain/profile for the default admin user?

- A. local
- B. internal
- C. default
- D. admin

38. If you enable automatic firewall configuration during engine-setup, what happens?

- A. All ports are opened
- B. The firewall is disabled
- C. Necessary ports for OLVM are automatically allowed
- D. Only SSH port is opened

ALTERNATE HOSTNAME CONFIGURATION

39. Where do you configure an alternate hostname for the OLVM Engine?

- A. /etc/hosts
- B. /etc/ovirt-engine/engine.conf
- C. /etc/ovirt-engine/engine.conf.d/custom-ss0-setup.conf
- D. /var/lib/ovirt-engine/hostname.conf

40. What command is used to restart the OLVM Engine service after configuring an alternate hostname?

- A. service ovirt-engine restart
- B. systemctl restart ovirt-engine
- C. engine-restart
- D. systemctl reload ovirt-engine

41. Which three benefits does configuring an alternate hostname provide? (Choose 3)

- A. Enhanced accessibility with multiple entry points
- B. Increased security
- C. Load balancing options
- D. Increased flexibility for network segmentation
- E. Faster VM performance

ADMINISTRATION PORTAL ACCESS

42. Which three portals are available in the OLVM Engine web interface? (Choose 3)

- A. Administration Portal
- B. Developer Portal
- C. VM Portal
- D. Monitoring Portal (Grafana)
- E. Storage Portal

43. What is the primary purpose of the Administration Portal?

- A. End users managing their VMs only
- B. Comprehensive management of all OLVM components by administrators
- C. Monitoring only
- D. Backup management

44. What is the Monitoring Portal in OLVM?

- A. A custom Oracle monitoring tool
- B. Grafana for monitoring and analytics
- C. A syslog server
- D. An SNMP trap receiver

45. After installing the Engine CA certificate in the browser, what does it allow you to do?

- A. Access VMs directly
- B. Securely access the OLVM Administration Portal without certificate warnings
- C. Increase browser performance
- D. Enable two-factor authentication

PRACTICAL IMPLEMENTATION

46. What command is used to enable a repository using dnf config-manager?

- A. `dnf config-manager enable <repo-name>`
- B. `dnf config-manager --set-enabled <repo-name>`
- C. `dnf enable-repo <repo-name>`
- D. `dnf repository enable <repo-name>`

47. What command cleans up the dnf repository cache?

- A. `dnf clear`
- B. `dnf clean`
- C. `dnf clean all`
- D. `dnf cache-clean`

48. After installing the ovirt-engine package, what is the FIRST command you run to begin configuration?

- A. `ovirt-config`
- B. `engine-setup`
- C. `systemctl start ovirt-engine`
- D. `engine-configure`

49. What file extension does the Engine CA certificate have when downloaded?

- A. `.crt`
- B. `.pem`
- C. `.cer`
- D. `.p12`

50. When installing the Engine CA certificate in a browser, which certificate store location is recommended?

- A. Current User
- B. Local Machine
- C. Trusted Root
- D. Personal

Answer Key — Engine Installation

1. B - Oracle Linux 8.5
2. A, C - Intel VT-x and AMD-V
3. B - 64-bit x86
4. A, B, C, D, E, F - All six listed
5. B - UEKR7
6. A, B, C - 2 core CPUs, 4 GB RAM, 25 GB local writable disk
7. B, C, E - 4 cores, 16 GB or greater RAM, 50 GB or greater local writable disk
8. A - 1 to 5 hosts
9. B - Up to 50 VMs
10. B, C, E - 8 core CPUs, 32 GB or greater RAM, 100 GB or greater local writable disk
11. B - 5 to 50 hosts
12. B - 50 to 500 VMs
13. B, C, E - 16 cores or greater, 64 GB or greater RAM, 200 GB or greater writable disk space
14. B - 50 to 200 hosts
15. B - 500 to 2000 VMs
16. B - 22
17. A, C - 80 (TCP) and 443 (TCP)
18. B - 5432
19. B, C - 54321 and 54322
20. C - 5900 to 6923
21. C - 3260
22. A - 111 (TCP and UDP)
23. B, C - pki-deps and postgresql:13
24. B - Public Key Infrastructure
25. B - virt:ol
26. C - virt:kvm_utils2
27. B - ol8_baseos_latest
28. C - dnf install oracle-ovirt-release-el8
29. B - dnf install ovirt-engine
30. B - engine-setup
31. C - Interactive command-line with questions
32. B - Summary of entered values
33. B - Details about how to log in to the Administration Portal
34. A, C, D, E, F - Database config, Network config, Admin user setup, Certificates/security, Service configs
35. A, B - A local database instance, A remote database
36. C - admin
37. B - internal
38. C - Necessary ports for OLVM are automatically allowed
39. C - /etc/ovirt-engine/engine.conf.d/custom-ss0-setup.conf
40. B - systemctl restart ovirt-engine
41. A, C, D - Enhanced accessibility, Load balancing, Increased flexibility

-
- 42. A, C, D - Administration Portal, VM Portal, Monitoring Portal
 - 43. B - Comprehensive management of all OLVM components by administrators
 - 44. B - Grafana for monitoring and analytics
 - 45. B - Securely access the OLVM Administration Portal without certificate warnings
 - 46. B - `dnf config-manager --set-enabled <repo-name>`
 - 47. C - `dnf clean all`
 - 48. B - `engine-setup`
 - 49. Not explicitly stated in transcript, but based on demo: likely .cer or certificate file
 - 50. B - Local Machine

Score Interpretation

- 46-50 correct: Excellent - Comprehensive mastery
- 41-45 correct: Very Good - Strong understanding
- 36-40 correct: Good - Ready with review
- 31-35 correct: Fair - Review weak areas
- Below 31: Needs comprehensive review

Section 3: Core Components

45 Questions

DATA CENTERS

1. What does a Data Center represent in Oracle Linux Virtualization Manager?

- A. A physical building location
- B. The highest organizational level encapsulating clusters, hosts, storage, and network configuration
- C. A type of storage domain
- D. A virtual machine template

2. Which four components does a Data Center encapsulate? (Choose 4)

- A. Clusters
- B. Applications
- C. Hosts
- D. Storage domains
- E. Network configuration
- F. User accounts

3. Can a single oVirt engine manage multiple Data Centers?

- A. No, only one Data Center per engine
- B. Yes, multiple Data Centers per engine
- C. Only in Self-Hosted Engine mode
- D. Only with special licensing

4. What is the maximum character limit for a Data Center name?

- A. 20 characters
- B. 30 characters
- C. 40 characters
- D. 50 characters

5. Which two storage types can be configured for a Data Center? (Choose 2)

- A. Local storage
- B. Cloud storage
- C. Shared storage
- D. Tape storage

6. Can local and shared storage domains be mixed in the same Data Center?

-
- A. Yes, they can be freely mixed
 - B. No, they cannot be mixed
 - C. Only in production environments
 - D. Only with Oracle support approval

7. Which storage protocols can be used for SHARED storage in a Data Center? (Choose 5)

- A. iSCSI
- B. SMB/CIFS
- C. NFS
- D. Fiber Channel (FC)
- E. POSIX
- F. Gluster File System

8. When would you typically use a local storage Data Center?

- A. Production enterprise environments
- B. Testing or development environments
- C. High-availability architectures
- D. Disaster recovery sites

9. What happens when a Data Center is configured with local storage? (Choose 3)

- A. It supports a single host only
- B. It enables high-availability features
- C. It creates a single non-shareable cluster
- D. Cluster management is not available
- E. VM migration is fully supported

10. Does a default Data Center get created during OLVM installation?

- A. Yes, one default Data Center is created
- B. No, you must manually create the first one
- C. Only in Self-Hosted Engine deployments
- D. Only if shared storage is detected

CLUSTERS

11. What is a Cluster in Oracle Linux Virtualization Manager?

- A. A physical rack of servers
- B. A group of hosts that share the same storage domains and network configuration
- C. A type of virtual machine
- D. A backup server group

12. What must all hosts within a cluster share? (Choose 2)

- A. The same CPU type
- B. The same RAM capacity

-
- C. The same storage domains
 - D. The same network configuration

13. Where do you access the Clusters management interface in the Administration Portal?

- A. Storage -> Clusters
- B. Compute -> Clusters
- C. Network -> Clusters
- D. Administration -> Clusters

14. What information is displayed in the Cluster list view? (Choose 2)

- A. Host count
- B. Administrator names
- C. VM count
- D. IP addresses

15. Does a default cluster get created during OLVM installation?

- A. Yes, in the default Data Center
- B. No, must be manually created
- C. Only for Self-Hosted Engine
- D. Only if hosts are detected

CLUSTER CONFIGURATION

16. Which CPU architecture types can be selected for a cluster? (Choose 3)

- A. x86-64
- B. ARM64
- C. ppc64
- D. s390x
- E. MIPS

17. What does the "Memory Overcommit" optimization feature allow?

- A. Using more physical memory than allocated
- B. Allocating more virtual memory than physical memory available
- C. Sharing memory between clusters
- D. Compressing VM memory

18. Which three scheduling policies can be configured for a cluster? (Choose 3)

- A. Even distribution
- B. Random distribution
- C. Power saving
- D. Cost optimization
- E. Strict (No overcommit)

19. What is the purpose of the Fencing Policy in a cluster?

- A. To encrypt network traffic
- B. To isolate and restart a non-responsive host
- C. To limit VM creation
- D. To manage storage quotas

20. What is the default time delay before fencing is initiated?

- A. 30 seconds
- B. 60 seconds
- C. 90 seconds
- D. 120 seconds

21. What do Migration Policies define?

- A. How VMs are created
- B. Storage allocation rules
- C. Thresholds for automatic VM migration to balance load and number of parallel migrations
- D. Network bandwidth limits

AFFINITY GROUPS & LABELS

22. What is the purpose of Positive Affinity rules for VMs?

- A. VMs are placed on different hosts
- B. VMs are placed on the same host
- C. VMs are suspended
- D. VMs are migrated frequently

23. What is the purpose of Negative Affinity rules for VMs?

- A. VMs are placed on the same host
- B. VMs are placed on different hosts
- C. VMs cannot be started
- D. VMs share the same storage

24. What does the Enforcement Mode setting determine for affinity rules?

- A. How many VMs can be created
- B. How strictly the affinity rules are enforced
- C. Storage allocation limits
- D. Network bandwidth

25. If the Enforcement Mode for an affinity rule is DISABLED, what happens?

- A. The rules are strictly enforced
- B. The rules are preferences but not strictly enforced
- C. The rules are ignored completely

D. VMs cannot be started

26. What are Affinity Labels used for?

- A. Encrypting VM data
- B. Tagging and categorizing VMs and hosts to group them logically
- C. Naming storage domains
- D. Configuring network VLANs

27. Which two types of affinity rules can be defined? (Choose 2)

- A. VM to VM affinity rules
- B. Storage to VM affinity rules
- C. VM to host affinity rules
- D. Network to host affinity rules

HOSTS

28. What are the two major types of physical hosts in OLVM? (Choose 2)

- A. Engine host
- B. Storage host
- C. Network host
- D. KVM host

29. What is the primary function of the Engine host?

- A. Running virtual machines
- B. Providing central management and administration tools
- C. Storing VM disk images
- D. Managing network switches

30. What is the primary function of KVM hosts?

- A. Managing the Administration Portal
- B. Hosting and running virtual machines
- C. Storing backup data
- D. Running the database

31. Can a single OLVM engine manage multiple KVM hosts?

- A. No, only one host per engine
- B. Yes, a single engine can manage multiple hosts
- C. Only in clustered mode
- D. Only with enterprise licensing

32. Where in the Administration Portal do you manage hosts?

- A. Storage -> Hosts
- B. Compute -> Hosts

-
- C. Network -> Hosts
 - D. Administration -> Hosts

33. What information is displayed in the host list view? (Choose 4)

- A. Host name or IP address
- B. Cluster name
- C. Data Center name
- D. Active virtual machines
- E. CPU temperature
- F. Memory used

34. What does the "Maintenance Mode" do for a host?

- A. Shuts down the host permanently
- B. Prepares the host for maintenance by migrating VMs away
- C. Increases host performance
- D. Resets host configuration

35. What is the purpose of converting a host to SPM (Storage Pool Manager) role?

- A. To increase VM performance
- B. To manage storage domains within the Data Center
- C. To enable backups
- D. To configure networks

STORAGE POOL MANAGER (SPM)

36. What is the Storage Pool Manager (SPM)?

- A. A physical storage device
- B. A management role assigned to one host in the Data Center
- C. A type of storage domain
- D. A backup application

37. How many hosts can have the SPM role in a single Data Center at one time?

- A. All hosts share the SPM role
- B. Only one host at a time
- C. Two hosts for redundancy
- D. Up to five hosts

38. What is the primary responsibility of the SPM?

- A. Running virtual machines
- B. Coordinating all metadata changes like creating disks or snapshots within the Data Center
- C. Managing network traffic
- D. User authentication

39. Does losing SPM functionality affect running VMs?

- A. Yes, all VMs stop immediately
- B. No, VMs continue running but storage operations are affected
- C. Only new VMs are affected
- D. VMs are automatically migrated

STORAGE DOMAINS

40. Which three types of storage domains exist in OLVM? (Choose 3)

- A. Data Domain
- B. Boot Domain
- C. ISO Domain
- D. Export Domain
- E. Template Domain

41. Can a storage domain be shared across multiple Data Centers?

- A. Yes, freely shared
- B. No, a storage domain belongs to only one Data Center
- C. Only ISO domains can be shared
- D. Only with special configuration

42. What must be configured for a Data Center before it can be initialized?

- A. At least one cluster
- B. At least one host
- C. At least one storage domain (Data Domain)
- D. At least one virtual machine

43. For VMs to be migrated between hosts, what storage requirement must be met?

- A. Each host needs local storage
- B. VMs must share the same storage domain
- C. Storage must be SSD-based
- D. VMs must have pre-allocated disks

VIRTUAL MACHINE STORAGE

44. What is the default provisioning method for virtual disks?

- A. Pre-allocated
- B. Thin provisioning
- C. Thick provisioning
- D. Compressed

45. What happens with thin-provisioned disks?

-
- A. Full disk space is allocated immediately
 - B. Disk space is allocated only as data is written, starting with minimal space
 - C. Disk cannot grow beyond initial size
 - D. Disk is stored on local storage only

46. When should pre-allocated disks be used?

- A. For test and development VMs
- B. For VMs with low I/O requirements
- C. For VMs with high levels of input/output operations
- D. Never, thin provisioning is always better

47. What benefit do pre-allocated disks provide for high I/O workloads?

- A. They use less storage space
- B. They maintain consistent performance by reducing latency from space allocation
- C. They enable faster VM migration
- D. They require less memory

EVENT LOGGING & NOTIFICATIONS

48. Which two primary log files are used in OLVM? (Choose 2)

- A. engine.log
- B. system.log
- C. vdsd.log
- D. kernel.log

49. Where is the engine log file typically located?

- A. /var/log/messages
- B. /var/log/ovirt-engine/engine.log
- C. /var/log/olvm/engine.log
- D. /etc/ovirt-engine/engine.log

50. What does the VDSM log file contain?

- A. User login activities
- B. Events and operations related to individual hosts including VM startups, migrations, and storage activities
- C. Network traffic logs
- D. Database queries

51. What tool is used to collect logs for sharing with Oracle support?

- A. logcollector
- B. ovirt-log-collector
- C. engine-backup
- D. sosreport

52. Where can administrators view alerts and events in the Administration Portal?

- A. Events tab
- B. Notification Drawer
- C. Dashboard only
- D. System Settings

53. Which two notification methods can be configured in OLVM? (Choose 2)

- A. Email notifications
- B. SMS messages
- C. SNMP traps
- D. Slack integration

Answer Key — Core Components

1. B - The highest organizational level encapsulating clusters, hosts, storage, and network configuration
2. A, C, D, E - Clusters, Hosts, Storage domains, Network configuration
3. B - Yes, multiple Data Centers per engine
4. C - 40 characters
5. A, C - Local storage, Shared storage
6. B - No, they cannot be mixed
7. A, C, D, E, F - iSCSI, NFS, FC, POSIX, Gluster FS
8. B - Testing or development environments
9. A, C, D - Single host only, Single non-shareable cluster, No cluster management
10. A - Yes, one default Data Center is created
11. B - A group of hosts that share the same storage domains and network configuration
12. C, D - Same storage domains, Same network configuration
13. B - Compute -> Clusters
14. A, C - Host count, VM count
15. A - Yes, in the default Data Center
16. A, C, D - x86-64, ppc64, s390x
17. B - Allocating more virtual memory than physical memory available
18. A, C, E - Even distribution, Power saving, Strict
19. B - To isolate and restart a non-responsive host
20. B - 60 seconds
21. C - Thresholds for automatic VM migration and parallel migrations
22. B - VMs are placed on the same host
23. B - VMs are placed on different hosts
24. B - How strictly the affinity rules are enforced
25. B - The rules are preferences but not strictly enforced
26. B - Tagging and categorizing VMs and hosts to group them logically
27. A, C - VM to VM affinity rules, VM to host affinity rules

-
- 28. A, D - Engine host, KVM host
 - 29. B - Providing central management and administration tools
 - 30. B - Hosting and running virtual machines
 - 31. B - Yes, a single engine can manage multiple hosts
 - 32. B - Compute -> Hosts
 - 33. A, B, C, D, F - Host name/IP, Cluster name, Data Center name, Active VMs, Memory used
 - 34. B - Prepares the host for maintenance by migrating VMs away
 - 35. B - To manage storage domains within the Data Center
 - 36. B - A management role assigned to one host in the Data Center
 - 37. B - Only one host at a time
 - 38. B - Coordinating all metadata changes like creating disks or snapshots
 - 39. B - No, VMs continue running but storage operations are affected
 - 40. A, C, D - Data Domain, ISO Domain, Export Domain
 - 41. B - No, a storage domain belongs to only one Data Center
 - 42. C - At least one storage domain (Data Domain)
 - 43. B - VMs must share the same storage domain
 - 44. B - Thin provisioning
 - 45. B - Disk space is allocated only as data is written, starting with minimal space
 - 46. C - For VMs with high levels of input/output operations
 - 47. B - They maintain consistent performance by reducing latency from space allocation
 - 48. A, C - engine.log, vdsd.log
 - 49. B - /var/log/ovirt-engine/engine.log
 - 50. B - Events and operations related to individual hosts including VM startups, migrations, and storage activities
 - 51. B - ovirt-log-collector
 - 52. B - Notification Drawer
 - 53. A, C - Email notifications, SNMP traps

Score Interpretation

- 48-53 correct: Excellent - Comprehensive mastery
- 43-47 correct: Very Good - Strong understanding
- 38-42 correct: Good - Ready with review
- 33-37 correct: Fair - Review weak areas
- Below 33: Needs comprehensive review

Key Exam Topics

- Data Center hierarchy and organization
- Storage type limitations (local vs shared, cannot mix)
- Cluster configuration (CPU types, policies, fencing)
- Affinity rules (positive vs negative, enforcement)
- SPM role (one per Data Center)
- Storage domains (one Data Center only, required before init)
- Thin vs pre-allocated disk provisioning

- Log file locations and log collection tools

Section 4: KVM Host Installation

50 Questions

KVM HOST PREREQUISITES

1. What is the minimum Oracle Linux version required for a KVM host?
 - A. Oracle Linux 7.5
 - B. Oracle Linux 8.5 or later
 - C. Oracle Linux 9.0
 - D. Oracle Linux 8.0
2. Which installation type is recommended for the KVM host operating system?
 - A. Full installation with GUI
 - B. Minimal install
 - C. Server with GUI
 - D. Workstation installation
3. Which two Unbreakable Enterprise Kernel releases are supported for KVM hosts? (Choose 2)
 - A. UEKR5
 - B. UEKR6
 - C. UEKR7
 - D. UEKR8
4. Which additional repository channel is required specifically for VDSM on the KVM host?
 - A. UEKR6
 - B. UEKR7
 - C. Gluster AppStream
 - D. KVM AppStream

CPU REQUIREMENTS

5. What is the MINIMUM CPU requirement for a KVM host?
 - A. Single-core 32-bit CPU
 - B. 64-bit dual-core CPU
 - C. 64-bit quad-core CPU
 - D. 64-bit eight-core CPU
6. What is RECOMMENDED for better performance in a KVM host?

-
- A. Single CPU with hyper-threading
 - B. Multiple CPUs
 - C. Higher clock speed only
 - D. Overclocked CPUs

7. Which two hardware virtualization technologies must the CPU support? (Choose 2)

- A. Intel VT-x
- B. Intel SGX
- C. AMD-V
- D. ARM TrustZone

MEMORY & STORAGE REQUIREMENTS

8. What is the MINIMUM RAM required for a KVM host?

- A. 1 GB
- B. 2 GB
- C. 4 GB
- D. 8 GB

9. What is the MAXIMUM tested RAM capacity for Oracle Linux Virtualization Manager hosts?

- A. 1 terabyte
- B. 2 terabytes
- C. 4 terabytes
- D. 6 terabytes

10. What is the minimum local storage requirement for a KVM host?

- A. 30 GB
- B. 40 GB
- C. 50 GB
- D. 60 GB

11. What is the purpose of the 60 GB local storage on a KVM host?

- A. Storing all virtual machine disks
- B. Dedicated for host operating system and OLVM components
- C. Backup storage only
- D. ISO image storage

NETWORK REQUIREMENTS

12. What is the MINIMUM network interface requirement for a KVM host?

- A. One NIC with 100 Mbps bandwidth
- B. One NIC with 1 Gbps bandwidth

-
- C. Two NICs with 1 Gbps bandwidth
 - D. Four NICs with 1 Gbps bandwidth

13. What is RECOMMENDED for network configuration on a KVM host?

- A. One NIC with 10 Gbps
- B. Two or more NICs with at least 1 Gbps bandwidth
- C. Four NICs with 100 Mbps
- D. Single bonded interface

14. Why are multiple NICs recommended for KVM hosts?

- A. For load balancing only
- B. For redundancy only
- C. NICs can be dedicated for network-intensive activities like VM migration
- D. To connect to multiple data centers

FIREWALL REQUIREMENTS

15. When are firewall configurations automatically set up on a KVM host?

- A. During OS installation
- B. When the host is registered with the engine
- C. Never, must be done manually
- D. Only for production hosts

16. What port is used for SSH access between the KVM host and Manager?

- A. 21
- B. 22
- C. 23
- D. 3389

17. Which two ports are used for web management and web access? (Choose 2)

- A. 8080
- B. 8000
- C. 8443
- D. 9000

18. What is the port range for VM Console Access using SPICE or VNC?

- A. 3389 to 3400
- B. 5800 to 5900
- C. 5900 to 6923
- D. 8000 to 9000

19. Which two ports are used for VDSM communication? (Choose 2)

- A. 54320

- B. 54321
- C. 54322
- D. 54323

20. Which three ports are required for high availability with configured cluster management? (Choose 3)

- A. 5405 UDP
- B. 5406 UDP
- C. 5407 TCP
- D. 2224 TCP
- E. 2225 UDP

SCALABILITY LIMITS

21. What is the MAXIMUM number of physical CPU cores tested for a KVM host?

- A. 128
- B. 256
- C. 384
- D. 512

22. What is the MAXIMUM tested memory capacity for a KVM host?

- A. 2 terabytes
- B. 4 terabytes
- C. 6 terabytes
- D. 8 terabytes

23. What is the MAXIMUM number of concurrently running VMs tested on a single KVM host?

- A. 300
- B. 400
- C. 500
- D. Above 600

REPOSITORY CONFIGURATION

24. Which repository contains the latest version of Oracle Linux 8 base OS packages?

- A. ol8_baseos
- B. ol8_baseos_latest
- C. ol8_latest
- D. baseos_latest

25. Which repository provides additional application packages not included in the base OS?

- A. ol8_applications

-
- B. ol8_AppStream
 - C. ol8_extras
 - D. ol8_apps

26. Which repository specifically provides KVM-related packages?

- A. ol8_virtualization
- B. ol8_kvm
- C. ol8_kvm_appstream
- D. kvm_utils

27. Which repository contains packages for oVirt 4.4 on Oracle Linux 8?

- A. ol8_ovirt
- B. ol8_ovirt44
- C. ovirt44_ol8
- D. ol8_virtualization44

28. Which repository provides supplementary packages for oVirt 4.4?

- A. ol8_ovirt44_additions
- B. ol8_ovirt44_extras
- C. ol8_ovirt_extras
- D. ovirt_supplemental

29. Which repository provides Gluster file system packages?

- A. ol8_gluster
- B. ol8_gluster_appstream
- C. ol8_64_gluster_appstream
- D. gluster_ol8

30. Which repository contains Unbreakable Enterprise Kernel Release 7 packages?

- A. ol8_UEKR7
- B. ol8_64_UEKR7
- C. UEKR7_ol8
- D. uek_release7

PREPARING KVM HOST

31. Which virt module must be DISABLED when preparing a KVM host?

- A. virt:rhel
- B. virt:ol
- C. virt:kvm
- D. virt:enterprise

32. Which virt module must be ENABLED when preparing a KVM host?

-
- A. virt:kvm
 - B. virt:kvm_utils
 - C. virt:kvm_utils2
 - D. virt:ol8

33. What command is used to install the Oracle oVirt release package on the KVM host?

- A. dnf install ovirt-release-el8
- B. dnf install oracle-ovirt-el8
- C. dnf install oracle-ovirt-release-el8
- D. dnf install olvm-release-el8

34. What command cleans the dnf repository cache after configuring repositories?

- A. dnf clear
- B. dnf clean
- C. dnf clean all
- D. dnf cache-clean

ADDING KVM HOST TO ENGINE

35. Where in the Administration Portal do you add a new KVM host?

- A. Storage -> Hosts
- B. Compute -> Hosts
- C. Network -> Hosts
- D. Configuration -> Hosts

36. What button do you click to begin adding a new host?

- A. Add
- B. New
- C. Create
- D. Register

37. Which three pieces of information are required when adding a new host? (Choose 3)

- A. Host name (unique identifier)
- B. Email address
- C. Hostname (FQDN)
- D. SSH port
- E. Physical location

38. Which two authentication methods can be used when adding a KVM host? (Choose 2)

- A. Password authentication
- B. Kerberos
- C. SSH key authentication
- D. Certificate authentication

39. For which user account must authentication credentials be provided when adding a host?

- A. admin user
- B. root user
- C. ovirt user
- D. vdsmd user

40. If you select SSH key authentication, where must the SSH public key be placed on the KVM host?

- A. /etc/ssh/authorized_keys
- B. /root/.ssh/authorized_keys
- C. /home/ovirt/.ssh/authorized_keys
- D. /var/lib/ovirt/authorized_keys

41. When adding multiple SSH keys to authorized_keys file, where should the engine's SSH key be placed?

- A. At the end of the file
- B. As the first key in the file
- C. Anywhere in the file
- D. In a separate file

42. What option in the Add Host wizard automatically configures firewall rules?

- A. Enable Firewall
- B. Configure Firewall
- C. Automatically Configure Host Firewall
- D. Setup Firewall Rules

43. After adding a host, what indicates the host is successfully registered and running?

- A. Red arrow icon
- B. Yellow arrow icon
- C. Green arrow icon
- D. Blue arrow icon

POST-INSTALLATION CONFIGURATION

44. Which command is used to set SELinux enforcing policies on the KVM host?

- A. selinux
- B. setenforce
- C. semanage
- D. sestatus

45. Which SELinux boolean policy allows fencing to connect to the network?

- A. fenced_network_connect

-
- B. fenced_can_network_connect
 - C. fence_network_enabled
 - D. network_fence_allow

46. Which SELinux boolean policy allows virtualization to use NFS?

- A. virt_nfs_enabled
- B. virt_use_nfs
- C. nfs_virt_enabled
- D. allow_virt_nfs

47. What is the purpose of setting the OTOPI_PYTHON environment variable?

- A. To enable Python on the host
- B. To specify the Python version for OTOPI operations and avoid package mismatches
- C. To install Python packages
- D. To configure Python security

48. Where is the OTOPI_PYTHON environment variable typically configured?

- A. /etc/python.env
- B. /etc/otopi.env.d/my-python.env
- C. /var/lib/otopi/python.conf
- D. /root/.python_env

49. In a testing/lab environment, what service might be disabled (though NOT recommended for production)?

- A. sshd
- B. NetworkManager
- C. firewalld
- D. vdsmd

50. What command is used to disable the firewall daemon?

- A. firewall-cmd --disable
- B. systemctl disable firewalld
- C. service firewalld disable
- D. firewall disable

Answer Key — KVM Host Installation

1. B - Oracle Linux 8.5 or later
2. B - Minimal install
3. B, C - UEKR6 and UEKR7
4. B - UEKR7

-
5. B - 64-bit dual-core CPU
 6. B - Multiple CPUs
 7. A, C - Intel VT-x and AMD-V
 8. B - 2 GB
 9. D - 6 terabytes
 10. D - 60 GB
 11. B - Dedicated for host operating system and OLVM components
 12. B - One NIC with 1 Gbps bandwidth
 13. B - Two or more NICs with at least 1 Gbps bandwidth
 14. C - NICs can be dedicated for network-intensive activities like VM migration
 15. B - When the host is registered with the engine
 16. B - 22
 17. A, C - 8080 and 8443
 18. C - 5900 to 6923
 19. B, C - 54321 and 54322
 20. A, B, D - 5405 UDP, 5406 UDP, 2224 TCP
 21. C - 384
 22. C - 6 terabytes
 23. D - Above 600
 24. B - ol8_baseos_latest
 25. B - ol8_AppStream
 26. C - ol8_kvm_appstream
 27. B - ol8_ovirt44
 28. B - ol8_ovirt44_extras
 29. C - ol8_64_gluster_appstream
 30. B - ol8_64_UEKR7
 31. B - virt:ol
 32. C - virt:kvm_utils2
 33. C - dnf install oracle-ovirt-release-el8
 34. C - dnf clean all
 35. B - Compute -> Hosts
 36. B - New
 37. A, C, D - Host name (unique), Hostname (FQDN), SSH port
 38. A, C - Password authentication and SSH key authentication
 39. B - root user
 40. B - /root/.ssh/authorized_keys
 41. B - As the first key in the file
 42. C - Automatically Configure Host Firewall
 43. C - Green arrow icon
 44. C - semanage
 45. B - fenced_can_network_connect
 46. B - virt_use_nfs
 47. B - To specify the Python version for OTOPI operations and avoid package mismatches

48. B - /etc/otopi.env.d/my-python.env

49. C - firewalld

50. B - systemctl disable firewalld

Score Interpretation

46-50 correct: Excellent - Comprehensive mastery

41-45 correct: Very Good - Strong understanding

36-40 correct: Good - Ready with review

31-35 correct: Fair - Review weak areas

Below 31: Needs comprehensive review

Key Exam Topics

- Hardware requirements (CPU, memory, storage, network)
- Scalability limits (384 CPUs, 6TB RAM, 600+ VMs)
- Repository names and purposes (7 repositories to know)
- Firewall port numbers (22, 8080/8443, 5900-6923, 54321/54322, 5405/5406/2224)
- virt module management (disable :ol, enable :kvm_utils2)
- SSH key authentication and authorized_keys placement
- SELinux policy configuration (fenced_can_network_connect, virt_use_nfs)
- Post-installation tasks (OTOPI_PYTHON, firewalld)

Section 5: Storage Domain Management

55 Questions

STORAGE OVERVIEW & FUNDAMENTALS

1. What does OLVM storage provide centralized storage for? (Choose 3)

- A. Virtual machine disk images
- B. User home directories
- C. ISO files
- D. Snapshots
- E. Email archives

2. Which storage protocol allows OLVM hosts to access block-level storage over TCP/IP networks?

- A. NFS
- B. iSCSI
- C. SMB
- D. HTTP

3. Which storage protocol allows OLVM hosts to connect to fiber channel SAN storage systems?

- A. NFS
- B. iSCSI
- C. FCP (Fiber Channel Protocol)
- D. GlusterFS

4. Which storage protocol is a distributed file system providing scalability and redundancy?

- A. NFS
- B. iSCSI
- C. FCP
- D. GlusterFS

5. Can a Data Center be initialized without a storage domain attached?

- A. Yes, storage is optional
- B. No, at least one storage domain must be attached before initialization
- C. Only in test environments
- D. Only for Self-Hosted Engine

STORAGE DOMAIN TYPES

6. Which type of storage does OLVM support? (Choose 2)

- A. Block device storage (SAN)
- B. Tape storage
- C. File system storage (NAS)
- D. Cloud object storage

7. Which two protocols are examples of block device (SAN) storage? (Choose 2)

- A. NFS
- B. iSCSI
- C. FCP
- D. GlusterFS

8. Which two protocols are examples of file system (NAS) storage? (Choose 2)

- A. iSCSI
- B. NFS
- C. FCP
- D. GlusterFS

9. For VMs to be migrated between hosts, what storage requirement must be met?

- A. Each host needs local storage
- B. VMs must share the same storage domain
- C. Storage must be SSD-based
- D. VMs must use iSCSI only

10. Can a storage domain be shared between multiple Data Centers?

- A. Yes, freely shared
- B. No, storage domains cannot be shared between different OLVM Data Centers
- C. Only ISO domains can be shared
- D. Only with special configuration

STORAGE POOL MANAGER (SPM)

11. What is the Storage Pool Manager (SPM)?

- A. A physical storage device
- B. A management role assigned to one host in the Data Center
- C. A type of storage domain
- D. A backup application

12. How many hosts can have the SPM role in a single Data Center at one time?

- A. All hosts share the SPM role equally
- B. Only one host at a time
- C. Two hosts for redundancy

D. Up to five hosts

13. What is the primary responsibility of the SPM?

- A. Running virtual machines
- B. Coordinating storage metadata operations like creating disks or snapshots
- C. Managing network traffic
- D. User authentication

14. If the host with the SPM role fails, what happens?

- A. All storage operations stop permanently
- B. OLVM automatically assigns the SPM role to another host
- C. The Data Center becomes unavailable
- D. All VMs must be restarted

LOCAL STORAGE DOMAIN

15. What type of cluster is created when configuring local storage?

- A. Multi-host cluster
- B. Single-host cluster
- C. High-availability cluster
- D. Distributed cluster

16. Can VMs in a local storage domain be migrated to other hosts?

- A. Yes, freely migrated
- B. No, migration is not possible
- C. Only with special configuration
- D. Only during maintenance windows

17. Can VMs in a local storage domain be fenced?

- A. Yes, fencing works normally
- B. No, fencing is not available
- C. Only manual fencing
- D. Only with power management

18. What happens when you configure a KVM host for local storage if it was previously in a clustered Data Center?

- A. It remains in the cluster
- B. It is automatically removed from the previous Data Center and cluster
- C. Managing network traffic
- C. It joins both clusters
- D. It requires manual removal first

19. What is the RECOMMENDED way to configure local storage?

-
- A. Use the root filesystem
 - B. Create a directory on existing partition
 - C. Use a separate logical volume or disk
 - D. Use network-attached storage

20. After creating and mounting a local storage disk, where should it be added to ensure automatic mounting on reboot?

- A. /etc/hosts
- B. /etc/fstab
- C. /etc/mount
- D. /etc/storage

21. What must the host's state be before configuring local storage?

- A. Up
- B. Maintenance
- C. Non-operational
- D. Installing

NFS STORAGE DOMAIN

22. What is the recommended operating system for OLVM manager and KVM hosts when using NFS storage?

- A. Any Linux distribution
- B. Oracle Linux
- C. Windows Server
- D. FreeBSD

23. What is the default user ID for the VDSM user?

- A. 0
- B. 36
- C. 100
- D. 1000

24. What is the default group ID for the KVM group?

- A. 0
- B. 36
- C. 100
- D. 1000

25. Which users are members of the KVM group? (Choose 2)

- A. root
- B. QEMU
- C. sanlock

D. vdsm

26. What command is used to install NFS utilities on the NFS server?

- A. yum install nfs
- B. yum install nfs-utils
- C. dnf install nfs-server
- D. yum install nfs-tools

27. Which services must be added to the firewall for NFS? (Choose 3)

- A. nfs
- B. ssh
- C. nfs3
- D. mountd
- E. rpc-bind

28. What command reloads the firewall after adding services?

- A. firewall-cmd restart
- B. firewall-cmd --reload
- C. systemctl restart firewalld
- D. service firewall reload

29. What must be the ownership of NFS share directories?

- A. root:root
- B. vds:m:kvm
- C. qemu:sanlock
- D. admin:users

30. What is the purpose of the /etc/exports file on an NFS server?

- A. To list exported users
- B. To define NFS export configurations for sharing directories
- C. To configure firewall rules
- D. To manage user permissions

31. What command applies NFS export configurations?

- A. exportfs -a
- B. exportfs -ra
- C. nfs-export apply
- D. systemctl reload nfs

iSCSI STORAGE DOMAIN

32. What must be installed on KVM hosts before using iSCSI storage?

- A. NFS utilities

-
- B. iSCSI initiator software
 - C. Fiber channel drivers
 - D. GlusterFS client

33. What information is required to discover iSCSI targets? (Choose 2)

- A. IP address of storage server
- B. MAC address of storage server
- C. Port number
- D. DNS hostname only

34. What is the default port number for iSCSI?

- A. 22
- B. 3260
- C. 5432
- D. 8080

35. What authentication method can optionally be used with iSCSI?

- A. Kerberos
- B. LDAP
- C. CHAP
- D. OAuth

36. What does LUN stand for in iSCSI terminology?

- A. Local Unit Number
- B. Logical Unit Number
- C. Linux Unified Node
- D. Link Universal Network

37. What is the purpose of iSCSI multipathing?

- A. To increase storage capacity
- B. To prevent host downtime caused by network path failures
- C. To encrypt data transfers
- D. To compress data

38. What is an iSCSI bond?

- A. A physical cable connection
- B. A logical connection that aggregates multiple physical and logical connections to iSCSI targets
- C. A storage encryption method
- D. A user authentication mechanism

39. How many paths to the same iSCSI target should be selected when configuring multipathing?

- A. Only one path
- B. All paths to the same target

-
- C. Maximum of two paths
 - D. Paths from different targets

GLUSTERFS STORAGE DOMAIN

40. What is GlusterFS?

- A. A block storage protocol
- B. A distributed file system
- C. A backup application
- D. A virtual machine hypervisor

41. What format is used to specify the GlusterFS volume when creating a storage domain?

- A. IP:/volume_name
- B. FQDN:/volume_name
- C. volume_name@FQDN
- D. //FQDN/volume_name

42. What port is typically used for GlusterFS management?

- A. 22
- B. 3260
- C. 24007
- D. The transcript does not specify

DETACHING STORAGE DOMAINS

43. What must be the state of a storage domain before it can be detached from a Data Center?

- A. Active
- B. Maintenance
- C. Locked
- D. Unattached

44. Can you move a storage domain to maintenance mode if a VM has a lease on it?

- A. Yes, always
- B. No, you cannot
- C. Only if the VM is powered off
- D. Only with admin privileges

45. Can you detach a storage domain from one Data Center and attach it to another?

- A. Yes, but not simultaneously to both
- B. No, once attached it's permanent
- C. Yes, it can be attached to multiple Data Centers
- D. Only ISO domains can be moved

46. What is the MINIMUM number of storage domains a Data Center must have to remain active?

- A. Zero
- B. One (at least one storage domain)
- C. Two
- D. Three

ISO IMAGE MANAGEMENT

47. Where do you upload ISO images in the Administration Portal?

- A. Compute -> VMs
- B. Storage -> Disks
- C. Network -> Images
- D. Administration -> Media

48. What button is used to begin uploading an ISO image?

- A. New
- B. Import
- C. Upload
- D. Add

49. What must be installed in the browser to successfully upload images without connection errors?

- A. Java plugin
- B. Flash player
- C. Engine CA certificate
- D. ActiveX controls

50. What error might occur if the transfer image client inactivity timeout is too low?

- A. Connection refused
- B. Timeout due to transfer inactivity error
- C. Disk full error
- D. Authentication failure

BLOCK DEVICE MANAGEMENT (OCI)

51. In the practice labs using Oracle Cloud Infrastructure, what type of attachment is used for block volumes?

- A. NFS
- B. iSCSI
- C. Paravirtualized
- D. SMB

52. What access mode is typically configured for block volumes attached to storage servers?

- A. Read-only
- B. Write-only
- C. Read-write
- D. No access

53. What command is used to list newly attached disks on a Linux system?

- A. lsblk
- B. fdisk -l
- C. df -h
- D. mount

54. After attaching block devices via iSCSI commands, what tool can partition the devices?

- A. parted
- B. fdisk
- C. gdisk
- D. All of the above

55. Where are pre-created block volumes located in Oracle Cloud Infrastructure?

- A. Compute -> Instances
- B. Storage -> Block Volumes
- C. Network -> Storage
- D. Database -> Volumes

Answer Key — Storage Domain Management

1. A, C, D - VM disk images, ISO files, Snapshots
2. B - iSCSI
3. C - FCP (Fiber Channel Protocol)
4. D - GlusterFS
5. B - No, at least one storage domain must be attached before initialization
6. A, C - Block device storage (SAN), File system storage (NAS)
7. B, C - iSCSI, FCP
8. B, D - NFS, GlusterFS
9. B - VMs must share the same storage domain
10. B - No, storage domains cannot be shared between different OLVM Data Centers
11. B - A management role assigned to one host in the Data Center
12. B - Only one host at a time
13. B - Coordinating storage metadata operations like creating disks or snapshots
14. B - OLVM automatically assigns the SPM role to another host
15. B - Single-host cluster

-
16. B - No, migration is not possible
 17. B - No, fencing is not available
 18. B - It is automatically removed from the previous Data Center and cluster
 19. C - Use a separate logical volume or disk
 20. B - /etc/fstab
 21. B - Maintenance
 22. B - Oracle Linux
 23. B - 36
 24. B - 36
 25. B, C - QEMU, sanlock
 26. B - yum install nfs-utils
 27. A, D, E - nfs, mountd, rpc-bind
 28. B - firewall-cmd --reload
 29. B - vds:m:kvm
 30. B - To define NFS export configurations for sharing directories
 31. B - exportfs -ra
 32. B - iSCSI initiator software
 33. A, C - IP address of storage server, Port number
 34. B - 3260
 35. C - CHAP
 36. B - Logical Unit Number
 37. B - To prevent host downtime caused by network path failures
 38. B - A logical connection that aggregates multiple physical and logical connections to iSCSI targets
 39. B - All paths to the same target
 40. B - A distributed file system
 41. B - FQDN:/volume_name
 42. D - The transcript does not specify (Actually 24007 but not in transcript)
 43. B - Maintenance
 44. B - No, you cannot
 45. A - Yes, but not simultaneously to both
 46. B - One (at least one storage domain)
 47. B - Storage -> Disks
 48. C - Upload
 49. C - Engine CA certificate
 50. B - Timeout due to transfer inactivity error
 51. B - iSCSI
 52. C - Read-write
 53. B - fdisk -l
 54. D - All of the above
 55. B - Storage -> Block Volumes

Score Interpretation

51-55 correct: Excellent - Comprehensive mastery

46-50 correct: Very Good - Strong understanding

41-45 correct: Good - Ready with review

36-40 correct: Fair - Review weak areas

Below 36: Needs comprehensive review

Key Exam Topics

- Storage cannot be shared between Data Centers
- SPM: one host per Data Center, automatically reassigned
- Local storage: single-host cluster, no migration/fencing
- VDSM user ID and KVM group ID: both 36
- NFS ownership: vdsd:kvm
- iSCSI default port: 3260
- iSCSI multipathing: all paths to same target
- GlusterFS format: FQDN:/volume_name
- Storage domain maintenance required before detach
- Minimum one storage domain per Data Center

Section 6: Network Management

40 Questions

NETWORKING RECOMMENDATIONS

1. What is network bonding?

- A. Connecting VMs to the network
- B. Combining multiple network interfaces into a single logical interface to increase throughput and provide redundancy
- C. Encrypting network traffic
- D. Creating virtual networks

2. Why is network bonding recommended especially on production hosts?

- A. To reduce costs
- B. To ensure high availability and fault tolerance
- C. To simplify configuration
- D. To increase security only

3. What is the purpose of using VLANs in OLVM?

- A. To increase network speed
- B. To segment network traffic logically for enhanced security and performance
- C. To reduce the number of physical switches needed
- D. To enable wireless connectivity

4. Which types of traffic can be separated using VLANs? (Choose 3)

- A. Management network traffic
- B. Email traffic
- C. Storage network traffic
- D. User traffic
- E. DNS queries

5. When adding a physical NIC with VLAN support, where should the VLAN be assigned?

- A. To the virtual machine
- B. Directly to the physical interface
- C. To the logical network only
- D. To the cluster

6. What does FQDN stand for?

-
- A. Fast Query Domain Name
 - B. Fully Qualified Domain Name
 - C. Forward Query DNS Name
 - D. File Query Domain Network

7. Which two types of name resolution must OLVM support? (Choose 2)

- A. Forward name resolution (domain name to IP)
- B. Lateral name resolution
- C. Reverse name resolution (IP to domain name)
- D. Broadcast name resolution

8. Where should DNS servers used for OLVM name resolution be hosted?

- A. As VMs inside the OLVM environment
- B. On the engine host
- C. Outside of the OLVM environment
- D. On each KVM host

9. Why should DNS servers be hosted outside the OLVM environment?

- A. To reduce costs
- B. To prevent dependency loops and ensure DNS availability even if the virtual environment encounters issues
- C. To increase DNS performance
- D. Because VMs cannot run DNS services

LOGICAL NETWORKS

10. What are logical networks in OLVM?

- A. Physical network cables
- B. Representations of network resources that provide connectivity for KVM virtual machines
- C. Virtual switches only
- D. Network security policies

11. At what level are network roles assigned to logical networks?

- A. Data Center level
- B. Cluster level
- C. Host level
- D. VM level

12. What is the purpose of a Management network role?

- A. Running VMs
- B. Handling management traffic between OLVM engine and KVM hosts for administrative tasks
- C. Providing internet access
- D. Storing backups

13. What is the purpose of a VM network role?

- A. Managing the engine
- B. Carrying traffic generated by virtual machines
- C. Live migration only
- D. Storage access

14. What is the purpose of a Display network role?

- A. Managing VMs
- B. Handling display traffic such as graphical display output of virtual machines
- C. Storing VM images
- D. Network monitoring

15. What is the purpose of a Migration network role?

- A. Moving physical servers
- B. Handling live migration traffic when VMs are moved between hosts without downtime
- C. Backup operations
- D. User authentication

16. What is the name of the default logical network automatically created during OLVM setup?

- A. default_network
- B. ovirtmgmt
- C. management_net
- D. cluster_network

17. What role is typically assigned to the ovirtmgmt network?

- A. Display role
- B. Management role
- C. Storage role
- D. Migration role

18. How many logical networks can be configured as the default route for a KVM host?

- A. Zero
- B. One
- C. Two for redundancy
- D. As many as needed

19. What happens if a KVM host loses connectivity to a network marked as "required"?

- A. Nothing, it continues normally
- B. The host will be considered non-operational
- C. Only VMs on that network stop
- D. The host reboots automatically

20. For VM networks, what is created on the host for each logical network?

-
- A. A VLAN tag
 - B. A bridge (virtual switch)
 - C. A firewall rule
 - D. A routing table

BRIDGE NETWORKS

21. What does a network bridge act as on a KVM host?

- A. A router
- B. A virtual switch connecting VMs to the physical network
- C. A firewall
- D. A load balancer

22. In bridge network configuration, what do VMs use to connect to bridges?

- A. Physical NICs directly
- B. VNICs (Virtual Network Interface Cards)
- C. USB adapters
- D. Wireless connections

23. Can a single VM have multiple VNICs connected to different bridges?

- A. No, only one VNIC per VM
- B. Yes, VMs can have multiple VNICs connected to different bridges
- C. Only in special configurations
- D. Only on physical hosts

24. How do VMs on different hosts in the same logical network communicate?

- A. They cannot communicate
- B. Through the physical network infrastructure that connects the hosts
- C. Only through the engine
- D. Only through storage domains

VLAN CONFIGURATION

25. What does VLAN stand for?

- A. Virtual LAN
- B. Virtual Local Area Network
- C. Verified LAN
- D. Variable Link Access Network

26. What is a VLAN identified by?

- A. MAC address
- B. IP address

-
- C. VLAN ID
 - D. Hostname

27. What device uses the VLAN ID to segregate traffic?

- A. The router
- B. The switch
- C. The firewall
- D. The DNS server

28. If a host does not have enough physical NICs, what can be used instead?

- A. Virtual NICs only
- B. VLAN configurations to create multiple virtual LANs
- C. Bonded interfaces only
- D. Additional virtual machines

29. Why must VLANs be configured on the physical network infrastructure before use in OLVM?

- A. To reduce costs
- B. To support logical networks that require VLANs for traffic segregation
- C. To increase security only
- D. To enable wireless access

VIRTUAL NICs (VNICs)

30. What does a VM use to connect to a logical network?

- A. Physical NIC directly
- B. VNIC (Virtual Network Interface Controller)
- C. USB adapter
- D. Serial port

31. What are VNICs always attached to on the KVM host?

- A. A VLAN
- B. A bridge
- C. A router
- D. A firewall

32. What does Oracle Linux Virtualization Manager automatically assign to a VNIC?

- A. IP address
- B. MAC address
- C. VLAN ID
- D. Hostname

33. What does each MAC address assigned to a VNIC correspond to?

- A. Multiple VNICs

-
- B. A single VNIC
 - C. All VNICs on a host
 - D. All VNICs in a cluster

34. What is the purpose of the MAC address on a VNIC?

- A. Encryption
- B. Identifying the VNIC on the network and ensuring proper communication
- C. Routing only
- D. Storage access

NETWORK BONDING

35. In a bonded network configuration, what are multiple NICs combined into?

- A. Multiple separate networks
- B. A single logical interface (bond)
- C. Virtual switches
- D. VLAN tags

36. After creating a bond, what is created over the bonded interface?

- A. A firewall
- B. A bridge
- C. A VM
- D. A storage domain

37. What benefit does bonding provide for network transmission?

- A. Reduces costs
- B. Combines transmission capabilities of all NICs for increased bandwidth and redundancy
- C. Simplifies configuration only
- D. Reduces power consumption

PASS-THROUGH NETWORKING & SR-IOV

38. What does pass-through networking allow?

- A. Encrypted connections only
- B. Network traffic to bypass the software network stack, providing direct access to physical NICs
- C. Faster VM creation
- D. Automatic IP assignment

39. What is a benefit of pass-through networking?

- A. Easier configuration
- B. Reduced latency and improved performance
- C. Lower cost

D. More VMs per host

40. What is a limitation of VMs with pass-through NICs?

- A. They use more memory
- B. They are less flexible in terms of live migration because specific physical hardware must be available on the target
- C. They cannot connect to networks
- D. They require more CPU cores

41. Which CPU features must be supported for pass-through networking? (Choose 2)

- A. Intel VT-X
- B. Intel VT-d
- C. AMD-V
- D. IOMMU support (Intel VT-d or AMD-Vi)

42. Can a physical NIC in pass-through mode be shared with other VMs on the same host?

- A. Yes, freely shared
- B. No, it cannot be shared while in pass-through mode
- C. Only with the same operating system
- D. Only in high-availability mode

43. What does SR-IOV stand for?

- A. Standard Root IO Virtualization
- B. Single Root IO Virtualization
- C. Shared Resource IO Virtualization
- D. Secure Root IO Virtualization

44. What does SR-IOV allow?

- A. Multiple operating systems on one VM
- B. A single physical network interface to be divided into multiple virtual interfaces (virtual functions)
- C. Faster storage access
- D. Encrypted networking

45. How do you display virtual functions associated with network interfaces in the Administration Portal?

- A. Click Enable VF
- B. Click Show Virtual Functions button
- C. Edit host properties
- D. Create new network

46. How do you associate a logical network with a virtual function?

- A. Edit the VM
- B. Drag and drop the logical network from unassigned to the virtual function

-
- C. Run a command line tool
 - D. Restart the host

VM NETWORK INTERFACE CONFIGURATION

47. Where in the Administration Portal do you configure VM network interfaces?

- A. Storage -> Networks
- B. Compute -> Virtual Machines -> Network Interfaces tab
- C. Network -> Interfaces
- D. Configuration -> NICs

48. What can you configure for a VM's network interface? (Choose 2)

- A. Custom MAC address
- B. CPU allocation
- C. MAC address from pool
- D. Memory size

49. After configuring a custom MAC address for a VM's VNIC, what must you do for the changes to take effect?

- A. Reboot the host
- B. Restart the engine
- C. Start or restart the virtual machine
- D. Reboot all hosts

50. In the VM details page, which tab shows network interface information?

- A. General
- B. Network Interfaces
- C. Hardware
- D. Settings

Answer Key — Network Management

1. B - Combining multiple network interfaces into a single logical interface for throughput and redundancy
2. B - To ensure high availability and fault tolerance
3. B - To segment network traffic logically for enhanced security and performance
4. A, C, D - Management traffic, Storage traffic, User traffic
5. B - Directly to the physical interface
6. B - Fully Qualified Domain Name
7. A, C - Forward name resolution, Reverse name resolution
8. C - Outside of the OLVM environment
9. B - To prevent dependency loops and ensure DNS availability

-
10. B - Representations of network resources that provide connectivity for KVM VMs
 11. B - Cluster level
 12. B - Handling management traffic between OLVM engine and KVM hosts
 13. B - Carrying traffic generated by virtual machines
 14. B - Handling display traffic such as graphical display output of VMs
 15. B - Handling live migration traffic when VMs are moved between hosts
 16. B - ovirtmgmt
 17. B - Management role
 18. B - One
 19. B - The host will be considered non-operational
 20. B - A bridge (virtual switch)
 21. B - A virtual switch connecting VMs to the physical network
 22. B - VNICs (Virtual Network Interface Cards)
 23. B - Yes, VMs can have multiple VNICs connected to different bridges
 24. B - Through the physical network infrastructure that connects the hosts
 25. B - Virtual Local Area Network
 26. C - VLAN ID
 27. B - The switch
 28. B - VLAN configurations to create multiple virtual LANs
 29. B - To support logical networks that require VLANs for traffic segregation
 30. B - VNIC (Virtual Network Interface Controller)
 31. B - A bridge
 32. B - MAC address
 33. B - A single VNIC
 34. B - Identifying the VNIC on the network and ensuring proper communication
 35. B - A single logical interface (bond)
 36. B - A bridge
 37. B - Combines transmission capabilities of all NICs for increased bandwidth and redundancy
 38. B - Network traffic to bypass the software network stack, providing direct access to physical NICs
 39. B - Reduced latency and improved performance
 40. B - They are less flexible in live migration because specific hardware must be available
 41. B, D - Intel VT-d, IOMMU support
 42. B - No, it cannot be shared while in pass-through mode
 43. B - Single Root IO Virtualization
 44. B - A single physical NIC to be divided into multiple virtual interfaces
 45. B - Click Show Virtual Functions button
 46. B - Drag and drop the logical network from unassigned to the virtual function
 47. B - Compute -> Virtual Machines -> Network Interfaces tab
 48. A, C - Custom MAC address, MAC address from pool
 49. C - Start or restart the virtual machine
 50. B - Network Interfaces

Score Interpretation

46-50 correct: Excellent - Comprehensive mastery

41-45 correct: Very Good - Strong understanding

36-40 correct: Good - Ready with review

31-35 correct: Fair - Review weak areas

Below 31: Needs comprehensive review

Key Exam Topics

- Network bonding: high availability and fault tolerance
- VLANs: traffic segmentation and security
- DNS: must be hosted outside OLVM environment
- Logical network roles: Management, VM, Display, Migration
- Default network: ovirtmgmt (management role)
- Required networks: host becomes non-operational if unavailable
- Bridges: created for each VM network on the host
- VNICS: automatically assigned MAC addresses
- One default route per KVM host
- Pass-through: cannot share NICs, reduced migration flexibility
- SR-IOV: virtual functions from single physical NIC
- Network configuration at cluster level

Section 7: User Management

50 Questions

ADMINISTRATOR TASKS

1. Which administrative task involves creating and managing user accounts with appropriate roles?

- A. Monitoring resources
- B. Managing user setup and setting permission levels
- C. Generating reports
- D. Managing physical resources

2. What are the main categories of resources that administrators manage in OLVM? (Choose 2)

- A. Physical servers
- B. Email systems
- C. Virtual resources (VMs, vCPUs, memory, storage, networks)
- D. Office applications

3. What should administrators set up for critical resource thresholds?

- A. Firewalls
- B. Alerts for critical thresholds and anomalies
- C. User accounts
- D. Virtual machines

4. What can administrators use to organize and manage OLVM objects?

- A. Folders
- B. Custom tags
- C. Directories
- D. Scripts

USER DOMAINS

5. What are the two types of user domains in OLVM? (Choose 2)

- A. Local domain
- B. Cloud domain
- C. External domain
- D. Hybrid domain

6. What is the name of the default local domain created during initial OLVM setup?

- A. local
- B. default
- C. internal
- D. admin

7. What is the default user created in the internal domain?

- A. root
- B. administrator
- C. admin
- D. superuser

8. When is the password for the default admin user set?

- A. After first login
- B. During engine setup
- C. Never, it's automatically generated
- D. When first VM is created

9. What tool is used to create additional users on the internal domain?

- A. useradd
- B. ovirt-user-tool
- C. ovirt-aaa-jdbc-tool
- D. engine-user-create

10. What are users created on local domains called?

- A. Internal users
- B. Local users
- C. Domain users
- D. System users

EXTERNAL DOMAINS

11. Which two external directory servers can be attached to OLVM? (Choose 2)

- A. Active Directory
- B. NIS
- C. OpenLDAP
- D. Kerberos

12. What are users created on external domains called?

- A. External users
- B. Directory users
- C. LDAP users

D. Remote users

13. What are the benefits of using external domains? (Choose 3)

- A. Centralized user management
- B. Lower costs
- C. Single sign-on capabilities
- D. Adherence to organization's security policies
- E. Faster VM performance

14. Where do directory users authenticate?

- A. Within OLVM's internal system
- B. Against their respective external directory servers
- C. At the hypervisor level
- D. Through the database

CREATING AND MANAGING USERS

15. What command is used to add a new user to the internal domain?

- A. ovirt-aaa-jdbc-tool add-user
- B. ovirt-aaa-jdbc-tool user add
- C. useradd
- D. engine-user-add

16. What parameter is used to set a user's password when creating a user?

- A. --pass
- B. --pwd
- C. --password
- D. --passwd

17. What command shows details of a specific user?

- A. ovirt-aaa-jdbc-tool user list
- B. ovirt-aaa-jdbc-tool user show <username>
- C. ovirt-aaa-jdbc-tool display user
- D. ovirt-aaa-jdbc-tool query user

18. What command is used to edit user attributes?

- A. ovirt-aaa-jdbc-tool user modify
- B. ovirt-aaa-jdbc-tool user change
- C. ovirt-aaa-jdbc-tool user edit
- D. ovirt-aaa-jdbc-tool update user

19. Which parameter is used to specify the attribute to change when editing a user?

- A. --parameter

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- B. --field
 - C. --attribute
 - D. --property

20. How do you disable a user account?

- A. ovirt-aaa-jdbc-tool user delete
- B. ovirt-aaa-jdbc-tool user disable
- C. ovirt-aaa-jdbc-tool user edit --flag=+disabled
- D. ovirt-aaa-jdbc-tool user lock

21. How do you re-enable a disabled user account?

- A. ovirt-aaa-jdbc-tool user enable
- B. ovirt-aaa-jdbc-tool user edit --flag=+enabled
- C. ovirt-aaa-jdbc-tool user unlock
- D. ovirt-aaa-jdbc-tool user activate

MANAGING GROUPS

22. What command is used to add a new group?

- A. ovirt-aaa-jdbc-tool add-group
- B. groupadd
- C. ovirt-aaa-jdbc-tool group add
- D. engine-group-create

23. What command is used to add users to a group?

- A. ovirt-aaa-jdbc-tool group adduser
- B. ovirt-aaa-jdbc-tool group-manage useradd
- C. ovirt-aaa-jdbc-tool user-to-group
- D. usermod -aG

24. When specifying multiple users to add to a group, how can they be separated? (Choose 2)

- A. With commas
- B. With semicolons
- C. With multiple --user flags separated by spaces
- D. With pipe symbols

25. What command lists all group manage options?

- A. ovirt-aaa-jdbc-tool group --help
- B. ovirt-aaa-jdbc-tool group-manage --help
- C. ovirt-aaa-jdbc-tool help groups
- D. man ovirt-groups

26. What command shows details of a specific group?

-
- A. ovirt-aaa-jdbc-tool group list
 - B. ovirt-aaa-jdbc-tool group display
 - C. ovirt-aaa-jdbc-tool group show <groupname>
 - D. ovirt-aaa-jdbc-tool query group

NESTED GROUPS

27. What are nested groups?

- A. Groups within VMs
- B. Hierarchical structures where groups can contain other groups (sub-groups) as members
- C. Encrypted group structures
- D. Groups spanning multiple data centers

28. How do you add a subgroup to a parent group?

- A. ovirt-aaa-jdbc-tool group add-subgroup
- B. ovirt-aaa-jdbc-tool group nest
- C. ovirt-aaa-jdbc-tool group-manage groupadd <parent-group> --group=<sub-group>
- D. ovirt-aaa-jdbc-tool subgroup create

29. What is a benefit of nested groups?

- A. Faster VM performance
- B. Reduced redundancy in permission assignments
- C. Lower storage costs
- D. Better network performance

QUERYING USERS AND GROUPS

30. What command is used to query user or group information?

- A. ovirt-aaa-jdbc-tool search
- B. ovirt-aaa-jdbc-tool find
- C. ovirt-aaa-jdbc-tool query
- D. ovirt-aaa-jdbc-tool list

31. What parameter specifies what to query (users or groups)?

- A. --type
- B. --what
- C. --object
- D. --target

32. What parameter specifies the search pattern when querying?

- A. --search
- B. --filter

-
- C. --pattern
 - D. --match

33. To query all users, what value is used with --what parameter?

- A. users
- B. user
- C. all-users
- D. userlist

34. To query all groups, what value is used with --what parameter?

- A. groups
- B. group
- C. all-groups
- D. grouplist

ACCOUNT SETTINGS

35. What command is used to view current account settings?

- A. ovirt-aaa-jdbc-tool settings list
- B. ovirt-aaa-jdbc-tool settings show
- C. ovirt-aaa-jdbc-tool show settings
- D. ovirt-aaa-jdbc-tool display-settings

36. What command is used to modify account settings?

- A. ovirt-aaa-jdbc-tool settings change
- B. ovirt-aaa-jdbc-tool settings modify
- C. ovirt-aaa-jdbc-tool settings set
- D. ovirt-aaa-jdbc-tool set settings

37. What is the default value for MAX_LOGIN_MINUTES?

- A. 60 minutes
- B. 1,440 minutes
- C. 10,080 minutes
- D. Unlimited

38. What setting controls the maximum number of failed login attempts before account lockout?

- A. MAX_FAILURES
- B. MAX_FAILURE_SINCE_SUCCESS
- C. LOGIN_ATTEMPTS_MAX
- D. FAILED_LOGIN_LIMIT

39. What is the default value for MAX_FAILURE_SINCE_SUCCESS?

- A. 3

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- B. 5
 - C. 10
 - D. Unlimited

40. What commands lock and unlock user accounts? (Choose 2)

- A. ovirt-aaa-jdbc-tool user lock <username>
- B. ovirt-aaa-jdbc-tool user unlock <username>
- C. ovirt-aaa-jdbc-tool user disable
- D. ovirt-aaa-jdbc-tool user freeze

MANAGING QUOTAS

41. What is a quota in OLVM?

- A. A backup limit
- B. A resource limitation tool to restrict memory, CPU, and storage usage
- C. A user count limit
- D. A network bandwidth limit

42. Which three types of resources can quotas limit? (Choose 3)

- A. Memory
- B. Email storage
- C. CPU
- D. Storage
- E. Network ports

43. What are the three quota modes in OLVM? (Choose 3)

- A. Enforced
- B. Monitored
- C. Audit
- D. Disabled
- E. Warning

44. What does the "Enforced" quota mode do?

- A. Logs violations only
- B. Puts into effect the quota limits that have been set
- C. Disables quotas
- D. Sends email alerts

45. What does the "Audit" quota mode do?

- A. Blocks all users
- B. Logs quota violations without blocking users
- C. Enforces strict limits
- D. Disables all quotas

46. What does the "Disabled" quota mode do?

- A. Enables strict quotas
- B. Logs violations
- C. Turns off runtime and storage limitations defined by quotas
- D. Sends warnings only

47. At what level is the quota mode configured?

- A. Cluster level
- B. Data Center level
- C. Host level
- D. VM level

48. What is a cluster threshold in quota settings?

- A. CPU speed limit
- B. The limit on total storage that can be allocated across all VMs within a specific cluster
- C. Network bandwidth limit
- D. Number of VMs allowed

49. What is cluster grace in quota settings?

- A. Extra time before enforcement
- B. The amount of cluster resources available after exhausting the cluster threshold
- C. Backup storage allocation
- D. Emergency CPU allocation

50. Where in the Administration Portal do you create and manage quota policies?

- A. Compute -> Quotas
- B. Storage -> Quotas
- C. Administration -> Quota
- D. Configuration -> Quotas

Answer Key — User Management

1. B - Managing user setup and setting permission levels
2. A, C - Physical servers, Virtual resources
3. B - Alerts for critical thresholds and anomalies
4. B - Custom tags
5. A, C - Local domain, External domain
6. C - internal
7. C - admin
8. B - During engine setup
9. C - ovirt-aaa-jdbc-tool

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10. B - Local users
 11. A, C - Active Directory, OpenLDAP
 12. B - Directory users
 13. A, C, D - Centralized management, Single sign-on, Security policies
 14. B - Against their respective external directory servers
 15. B - ovirt-aaa-jdbc-tool user add
 16. C - --password
 17. B - ovirt-aaa-jdbc-tool user show <username>
 18. C - ovirt-aaa-jdbc-tool user edit
 19. C - --attribute
 20. C - ovirt-aaa-jdbc-tool user edit --flag=+disabled
 21. B - ovirt-aaa-jdbc-tool user edit --flag=+enabled
 22. C - ovirt-aaa-jdbc-tool group add
 23. B - ovirt-aaa-jdbc-tool group-manage useradd
 24. A, C - With commas, With multiple --user flags
 25. B - ovirt-aaa-jdbc-tool group-manage --help
 26. C - ovirt-aaa-jdbc-tool group show <groupname>
 27. B - Hierarchical structures where groups contain other groups
 28. C - ovirt-aaa-jdbc-tool group-manage groupadd <parent> --group=<sub>
 29. B - Reduced redundancy in permission assignments
 30. C - ovirt-aaa-jdbc-tool query
 31. B - --what
 32. C - --pattern
 33. B - user
 34. B - group
 35. B - ovirt-aaa-jdbc-tool settings show
 36. C - ovirt-aaa-jdbc-tool settings set
 37. C - 10,080 minutes
 38. B - MAX_FAILURE_SINCE_SUCCESS
 39. B - 5
 40. A, B - user lock, user unlock
 41. B - A resource limitation tool
 42. A, C, D - Memory, CPU, Storage
 43. A, C, D - Enforced, Audit, Disabled
 44. B - Puts into effect the quota limits
 45. B - Logs quota violations without blocking users
 46. C - Turns off runtime and storage limitations
 47. B - Data Center level
 48. B - Limit on total storage across all VMs in cluster
 49. B - Amount of cluster resources available after exhausting threshold
 50. C - Administration -> Quota

Score Interpretation

46-50 correct: Excellent - Comprehensive mastery
41-45 correct: Very Good - Strong understanding
36-40 correct: Good - Ready with review
31-35 correct: Fair - Review weak areas
Below 31: Needs comprehensive review

Key Exam Topics

- Two domain types: local (internal) and external (AD/LDAP)
- Default domain: internal, default user: admin
- ovirt-aaa-jdbc-tool: primary command for user/group management
- User commands: user add, user edit, user show
- Group commands: group add, group-manage useradd
- Nested groups: hierarchical group structures
- Query command: --what and --pattern parameters
- Account settings: MAX_LOGIN_MINUTES (10080), MAX_FAILURE_SINCE_SUCCESS (5)
- Lock/unlock: user lock, user unlock
- Three quota modes: Enforced, Audit, Disabled
- Quota types: memory, CPU, storage
- Quota configuration: Data Center level
- Cluster threshold vs cluster grace

Section 8: Optimization Practices

45 Questions

MEMORY OVERCOMMITMENT

1. What does memory overcommitment in OLVM allow?

- A. Using only physical memory
- B. Total memory allocated to VMs can exceed physical memory capacity of the host
- C. Reducing VM density
- D. Disabling memory management

2. What is a benefit of memory overcommitment?

- A. Lower costs only
- B. Higher VM densities per host, maximizing resource efficiency
- C. Simpler configuration
- D. Reduced security

3. What can excessive memory overcommitment lead to?

- A. Faster performance
- B. Performance degradation if not managed effectively
- C. Increased security
- D. Lower VM density

4. What happens with higher overcommit threshold values?

- A. They increase memory usage
- B. They conserve physical memory but may increase CPU usage
- C. They disable overcommitment
- D. They reduce CPU usage

5. What does MoM stand for in OLVM?

- A. Mode of Management
- B. Memory Overcommitment Manager
- C. Monitor of Memory
- D. Management of Modules

6. What is KSM in the context of memory management?

- A. Key Storage Manager
- B. Kernel Same-page Merging

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- C. KVM System Memory
 - D. Kernel Security Module

7. What does KSM do?

- A. Encrypts memory pages
- B. Identifies identical memory pages across VMs and merges them into a single copy
- C. Increases memory allocation
- D. Backs up memory to storage

CPU OVERCOMMITMENT

8. Can OLVM hosts allocate more virtual CPUs to VMs than physical CPU cores available?

- A. No, never allowed
- B. Yes, vCPUs can exceed physical CPU core count
- C. Only with special licensing
- D. Only in test environments

9. What ensures fair CPU allocation among VMs on the same host?

- A. Manual intervention
- B. OLVM CPU schedulers
- C. Guest OS schedulers only
- D. Hardware only

RESOURCE ALLOCATION

10. How can administrators set CPU limits for VMs?

- A. Only by rebooting
- B. In terms of percentage of total host CPU resources or specific CPU cycles
- C. Through guest OS only
- D. Cannot be set

11. What is CPU pinning?

- A. Limiting CPU usage
- B. Assigning specific vCPUs to dedicated physical CPU cores or threads on the host
- C. Disabling CPUs
- D. Sharing CPUs among VMs

12. What is a benefit of CPU pinning?

- A. Lower costs
- B. Optimizes performance by reducing cache misses and ensuring consistent access
- C. Simpler configuration
- D. Higher VM density

13. What is memory reservation?

- A. Blocking memory access
- B. Guaranteeing a minimum amount of memory always available to a VM
- C. Encrypting memory
- D. Reducing memory allocation

14. What does the memory balloon device allow?

- A. Increasing physical memory
- B. VMs to reclaim unused memory and return it to the host for other VMs
- C. Encrypting memory
- D. Blocking memory access

15. Where is memory balloon optimization configured?

- A. VM level only
- B. Cluster level
- C. Data Center level
- D. Host level only

16. What does VirtIO disk optimization do?

- A. Encrypts disks
- B. Pins VirtIO disks to dedicated threads or CPU cores to reduce I/O contention
- C. Increases disk size
- D. Backs up disks

CLUSTER OPTIMIZATION & BALLOONING

17. What is ballooning in memory management?

- A. Expanding physical memory
- B. A technique where a balloon driver requests memory from guest OS to reduce VM memory usage
- C. Encrypting memory
- D. Backing up memory

18. Is the balloon device included in VMs by default?

- A. No, must be manually added
- B. Yes, unless explicitly removed during VM configuration
- C. Only in Linux VMs
- D. Only in Windows VMs

19. What must be installed in the guest OS for the balloon device to work?

- A. Additional storage drivers
- B. Relevant balloon drivers
- C. Network drivers

D. Video drivers

MEMORY OVERCOMMIT SETTINGS

20. What command is used to check current memory overcommit settings?

- A. olvm-config
- B. engine-config
- C. memory-check
- D. vm-config

21. What parameter can you grep to view memory overcommit settings?

- A. maxmem
- B. maxvdsmem
- C. memorymax
- D. vdsmemory

22. Which two parameters define memory overcommit ratios?

- A. MaxMemory and MinMemory
- B. MaxVdsMemOverCommit and MaxVdsMemOverCommitForServers
- C. MemoryLimit and MemoryReserve
- D. VMMemory and HostMemory

23. What command is used to change memory overcommit settings?

- A. engine-config -s
- B. olvm-set-memory
- C. vm-memory-config
- D. set-overcommit

HIGH AVAILABILITY & FENCING

24. Where should a fencing proxy host be located?

- A. In a different data center
- B. In the same cluster or data center as the host requiring fencing
- C. On the engine host only
- D. Outside the OLVM environment

25. Which three methods can perform power management operations? (Choose 3)

- A. By the manager after it reboots
- B. By the proxy host
- C. Manually in the administration portal
- D. By the guest OS
- E. By the storage system

26. What does OLVM automatically do to non-responsive hosts with power management enabled?

- A. Deletes them
- B. Attempts to fence them
- C. Ignores them
- D. Reboots all VMs

27. Why might a host appear non-responsive during boot?

- A. Hardware failure only
- B. Temporary delays due to initialization routines or startup tasks
- C. Network attacks
- D. Storage corruption

28. What command parameter prevents fencing during host bootstrap?

- A. disable-fencing
- B. DisableFenceAtStartupInSeconds
- C. no-fence-boot
- D. boot-fence-disable

29. What does PMHealthCheckEnabled do when set to true?

- A. Disables health checks
- B. Initiates regular health checks for all configured host power management agents
- C. Only checks once
- D. Checks only failed hosts

30. What does PMHealthCheckIntervalInSeconds specify?

- A. Boot timeout
- B. The interval at which OLVM performs health checks for power management agents
- C. VM restart time
- D. Storage check frequency

VM MIGRATION & HIGH AVAILABILITY

31. When does a highly available VM automatically restart or migrate? (Choose 2)

- A. During normal operation
- B. When its original host experiences failure
- C. When scheduled for maintenance
- D. When VM is idle
- E. When storage is full

32. What happens when a host is placed into maintenance mode?

- A. VMs are deleted
- B. VMs are gracefully shut down or migrated to another host before maintenance

- C. VMs continue running on that host
- D. All VMs are paused

33. For VM migration, what must both source and destination hosts access? (Choose 2)

- A. The same data domain where VM disks reside
- B. Different data domains
- C. The same virtual networks and VLANs
- D. Different network segments

34. What is required for highly available VMs?

- A. Single host only
- B. Host must be part of a cluster of two or more available hosts
- C. Local storage only
- D. Shared CPU only

HUGE PAGES

35. What is the typical size range for huge pages?

- A. 4 KB to 8 KB
- B. 2 MB to 1 GB
- C. 1 KB to 2 KB
- D. 4 GB to 8 GB

36. What does using huge pages reduce?

- A. Memory capacity
- B. The number of entries required in the system's page table
- C. CPU cores
- D. Storage space

37. What is TLB in the context of huge pages?

- A. Total Logical Buffer
- B. Translation Lookaside Buffer - hardware cache for virtual to physical address translations
- C. Thread Load Balancer
- D. Time Limit Buffer

38. How do larger page sizes affect TLB?

- A. Increase TLB misses
- B. Reduce TLB misses because fewer translations are needed
- C. Disable TLB
- D. Have no effect

39. When are huge pages allocated?

- A. Dynamically during runtime

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- B. Pre-allocated when a VM starts
 - C. Only when needed
 - D. After VM is running for 1 hour

40. What is a limitation of huge pages?

- A. Cannot use with Linux VMs
- B. Do not support hotplug or hot unplug operations for memory
- C. Reduce performance
- D. Increase memory usage

HOT PLUGGING vCPUs

41. What is hot plugging?

- A. Restarting VMs
- B. Adding or removing vCPUs from a running VM without downtime
- C. Increasing temperature
- D. Adding physical CPUs

42. Can you hot unplug a vCPU that was part of the original VM configuration?

- A. Yes, always
- B. No, only vCPUs that were previously hot plugged can be hot unplugged
- C. Only on Linux VMs
- D. Only with engine restart

43. What is the minimum number of vCPUs a VM cannot go below?

- A. 1 vCPU
- B. The number it was originally created with
- C. 2 vCPUs
- D. 4 vCPUs

44. What is the MINIMUM number of vCPUs required for hot plugging capability?

- A. 1 vCPU
- B. 2 vCPUs
- C. 4 vCPUs
- D. 8 vCPUs

45. What must Windows VMs have installed for hot plugging to work?

- A. Additional drivers only
- B. Guest agent
- C. Service packs
- D. .NET Framework

46. What must the VM's operating system support for hot plugging?

-
- A. Any OS works
 - B. Must explicitly support CPU hot plug functionality
 - C. Only Windows is supported
 - D. Only Linux is supported

HOT PLUGGING MEMORY

47. What happens each time you hot plug memory to a VM?

- A. VM restarts
- B. It appears as a new memory device under VM devices
- C. Old memory is removed
- D. VM is paused

48. What happens to hot plugged memory devices when you shut down and restart a VM?

- A. They become permanent
- B. They are cleared from VM devices without reducing the virtual memory space
- C. Memory is lost
- D. VM cannot restart

49. Can memory be hot plugged while a VM is running?

- A. No, VM must be stopped
- B. Yes, memory can be adjusted on the fly
- C. Only during maintenance mode
- D. Only once per day

50. What is incompatible with memory hot plug/unplug operations?

- A. VirtIO disks
- B. Huge pages
- C. Balloon devices
- D. Network interfaces

Answer Key — Optimization Practices

1. B - Total memory allocated to VMs can exceed physical memory capacity
2. B - Higher VM densities per host, maximizing resource efficiency
3. B - Performance degradation if not managed effectively
4. B - They conserve physical memory but may increase CPU usage
5. B - Memory Overcommitment Manager
6. B - Kernel Same-page Merging
7. B - Identifies identical memory pages and merges them
8. B - Yes, vCPUs can exceed physical CPU core count

-
9. B - OLVM CPU schedulers
 10. B - Percentage of total host CPU resources or specific CPU cycles
 11. B - Assigning specific vCPUs to dedicated physical CPU cores
 12. B - Optimizes performance by reducing cache misses
 13. B - Guaranteeing minimum memory always available to VM
 14. B - VMs reclaim unused memory and return to host
 15. B - Cluster level
 16. B - Pins VirtIO disks to dedicated threads to reduce I/O contention
 17. B - Balloon driver requests memory from guest OS to reduce VM memory
 18. B - Yes, unless explicitly removed during VM configuration
 19. B - Relevant balloon drivers
 20. B - engine-config
 21. B - maxvdsmem
 22. B - MaxVdsMemOverCommit and MaxVdsMemOverCommitForServers
 23. A - engine-config -s
 24. B - Same cluster or data center as host requiring fencing
 25. A, B, C - By manager after reboot, By proxy host, Manually in portal
 26. B - Attempts to fence them
 27. B - Temporary delays due to initialization routines
 28. B - DisableFenceAtStartupInSeconds
 29. B - Initiates regular health checks for power management agents
 30. B - Interval at which health checks are performed
 31. B, C - Host failure, Scheduled maintenance
 32. B - VMs gracefully shut down or migrated before maintenance
 33. A, C - Same data domain, Same virtual networks/VLANs
 34. B - Part of cluster of two or more available hosts
 35. B - 2 MB to 1 GB
 36. B - Number of entries in system's page table
 37. B - Translation Lookaside Buffer
 38. B - Reduce TLB misses
 39. B - Pre-allocated when VM starts
 40. B - Do not support hotplug/hot unplug for memory
 41. B - Adding/removing vCPUs from running VM without downtime
 42. B - No, only previously hot plugged vCPUs can be unplugged
 43. B - The number it was originally created with
 44. C - 4 vCPUs
 45. B - Guest agent
 46. B - Must explicitly support CPU hot plug functionality
 47. B - Appears as new memory device under VM devices
 48. B - Cleared from VM devices without reducing virtual memory space
 49. B - Yes, memory can be adjusted on the fly
 50. B - Huge pages

Score Interpretation

46-50 correct: Excellent - Comprehensive mastery

41-45 correct: Very Good - Strong understanding

36-40 correct: Good - Ready with review

31-35 correct: Fair - Review weak areas

Below 31: Needs comprehensive review

Key Exam Topics

- Memory overcommitment: higher VM density, potential performance issues
- MoM (Memory Overcommitment Manager) and KSM (Kernel Same-page Merging)
- CPU overcommitment: vCPUs can exceed physical cores
- CPU pinning: reduces cache misses, good for latency-sensitive apps
- Memory reservation: guarantees minimum memory
- Balloon device: included by default, requires guest drivers
- engine-config command: check/set overcommit (maxvdsmem)
- Fencing: same cluster/data center, DisableFenceAtStartupInSeconds
- VM migration: requires same data domain and virtual networks
- Huge pages: 2MB-1GB, reduce TLB misses, NO hotplug support
- vCPU hotplug: minimum 4 vCPUs, cannot go below original count
- Memory hotplug: appears as new device, cleared on restart
- Windows VMs: require guest agent for hotplug

Section 9: Virtual Machine Administration

45 Questions

VM STORAGE OVERVIEW

1. What are the two main types of virtual machine storage provisioning in OLVM? (Choose 2)

- A. Pre-allocated (thick provisioning)
- B. Compressed provisioning
- C. Sparse (thin provisioning)
- D. Encrypted provisioning

2. What is pre-allocated storage also known as?

- A. Thin provisioning
- B. Thick provisioning
- C. Sparse provisioning
- D. Dynamic provisioning

3. How does sparse allocation work?

- A. Allocates all storage upfront
- B. Defines maximum capacity but physical space is allocated only as data is written
- C. Compresses all data
- D. Encrypts all data

4. Which two disk format types does OLVM use? (Choose 2)

- A. QCOW2
- B. VMDK
- C. Raw
- D. VHD

5. What does QCOW2 stand for?

- A. Quality Copy On Write 2
- B. QEMU Copy On Write version 2
- C. Quick Copy Overwrite 2
- D. Queue Copy On Write 2

6. Which disk format supports snapshots, encryption, compression, and sparse allocation?

- A. Raw
- B. VMDK

-
- C. QCOW2
 - D. VHD

7. On NFS storage, what format is a QCOW2 disk created as?

- A. A block device
- B. A file with initial size close to 0
- C. A compressed archive
- D. An encrypted container

8. On SAN storage, what format is a QCOW2 disk created as?

- A. A file
- B. A block device with initial size smaller than defined virtual disk size
- C. A compressed file
- D. An LVM partition

9. On NFS storage, can raw disks be sparse?

- A. No, only pre-allocated
- B. Yes, they can be sparse files or pre-allocated
- C. Only with special configuration
- D. Raw disks are not supported on NFS

10. On SAN storage, are raw disks sparse or pre-allocated?

- A. Only sparse
- B. Both sparse and pre-allocated
- C. Only pre-allocated
- D. Neither, they use compression

WIPE AFTER DELETE

11. What does the "wipe after delete" flag do?

- A. Encrypts the disk
- B. Replaces user data with zeros when a virtual disk is deleted
- C. Backs up the disk
- D. Compresses the disk

12. On file storage systems like NFS, what effect does enabling "wipe after delete" have?

- A. It wipes data thoroughly
- B. It does nothing because the file system ensures no data exists
- C. It encrypts the data
- D. It backs up the data

13. When is enabling "wipe_after_delete" recommended?

- A. Always

-
- B. Never
 - C. If the virtual disk contained sensitive data
 - D. Only on SAN storage

14. What is the default value for the wipe_after_delete flag?

- A. True
- B. False
- C. Enabled
- D. Auto

15. What command is used to set SANWipeAfterDelete to true?

- A. olvm-config --set SANWipeAfterDelete=true
- B. engine-config --set SANWipeAfterDelete=true
- C. vm-config SANWipeAfterDelete true
- D. set-san-wipe true

16. After changing the SANWipeAfterDelete setting, what must you do?

- A. Reboot all hosts
- B. Restart the ovirt-engine service
- C. Restart all VMs
- D. Nothing, it takes effect immediately

17. What is a side effect of enabling wipe_after_delete?

- A. Faster deletion
- B. Performance degradation and prolonged delete times
- C. Increased storage capacity
- D. Better VM performance

SHAREABLE DISKS

18. What are shareable disks in OLVM?

- A. Disks shared between data centers
- B. Virtual disks that can be accessed simultaneously by multiple VMs
- C. Backup disks
- D. Network disks

19. What type of VMs should use shareable disks?

- A. Any VMs
- B. Only cluster-aware VMs
- C. Single VMs only
- D. Test VMs only

20. What can happen if you attach a shared disk to VMs that are not cluster-aware?

-
- A. Performance improvement
 - B. Data corruption
 - C. Increased security
 - D. Nothing

21. What is an appropriate use case for shareable disks?

- A. General file storage
- B. Clustered database servers or highly available services
- C. Single user workstations
- D. Test environments

CREATING VIRTUAL MACHINES

22. Where in the Administration Portal do you create a new VM?

- A. Storage -> VMs
- B. Compute -> Virtual Machines
- C. Network -> VMs
- D. Administration -> VMs

23. When creating a VM from scratch (not a template), what template option do you select?

- A. Default
- B. Blank
- C. Empty
- D. None

24. What are the two disk allocation policies? (Choose 2)

- A. Pre-allocated
- B. Compressed
- C. Thin provisioning (sparse)
- D. Encrypted

25. What is the recommended disk interface for VM disks?

- A. IDE
- B. SATA
- C. VirtIO-SCSI
- D. USB

26. In boot options, which device is typically selected for first boot when installing an OS?

- A. Hard disk
- B. CD-ROM
- C. Network
- D. USB

27. After OS installation is complete, which device should be the first boot device?

- A. CD-ROM
- B. Hard disk
- C. Network
- D. Floppy

28. What key do you press to access the boot menu in the VM console?

- A. F2
- B. Delete
- C. Escape
- D. F12

29. What key combination releases the cursor from the VM console?

- A. Ctrl+Alt
- B. Alt+F4
- C. Shift+F12
- D. Ctrl+Esc

VM SNAPSHOTS

30. What is a VM snapshot?

- A. A backup copy stored externally
- B. A point-in-time capture of VM state including memory, disk, and device state
- C. A template
- D. A compressed VM image

31. Should snapshots be used as a primary backup method?

- A. Yes, they are perfect for backups
- B. No, they should not be used as primary backup
- C. Only for production VMs
- D. Only for test VMs

32. Why should snapshots NOT be used as primary backup?

- A. They are too slow
- B. They are stored on the same storage as VM disks; if storage fails, snapshots become inaccessible
- C. They use too much CPU
- D. They are encrypted

33. When is it appropriate to take a snapshot?

- A. Every day as backup
- B. Before testing updates, configuration changes, or application patches
- C. Never

D. Only once per VM

34. What should you do with snapshots after reverting or when no longer needed?

- A. Keep them forever
- B. Delete them promptly to maintain optimal VM performance
- C. Archive them
- D. Encrypt them

35. What can happen if you take several snapshots without cleanup?

- A. Performance improvement
- B. Increased storage usage and performance degradation
- C. Better security
- D. Faster backups

36. What should be installed before taking snapshots?

- A. Antivirus software
- B. Latest guest agent package (qemu-guest-agent)
- C. Backup software
- D. Monitoring tools

37. Where do you create a VM snapshot in the Administration Portal?

- A. Storage -> Snapshots
- B. Compute -> Virtual Machines -> select VM -> Create Snapshot button
- C. Network -> Snapshots
- D. Configuration -> Snapshots

38. What can you optionally include when creating a snapshot?

- A. Network configuration
- B. Memory state of the VM
- C. User accounts
- D. Firewall rules

39. What happens to the VM when you select "Save Memory" during snapshot creation?

- A. VM continues running normally
- B. VM will be paused while saving memory
- C. VM is shut down
- D. VM is restarted

VM TEMPLATES

40. What is a VM template in OLVM?

- A. A backup file
- B. A pre-configured VM that captures configuration and state for quick deployment of new VMs

-
- C. A snapshot
 - D. A disk image

41. How do you create a template from a snapshot?

- A. Copy the VM
- B. Select snapshot -> click "Make Template" button
- C. Export the VM
- D. Clone the snapshot

42. Where in the Administration Portal do you view templates?

- A. Storage -> Templates
- B. Compute -> Templates
- C. Network -> Templates
- D. VM -> Templates

43. What option allows all users to access a template?

- A. Public template
- B. Allow all users to access this template
- C. Shared template
- D. Global template

44. What does "Seal template" do and for which OS is it available?

- A. Encrypts template, Windows only
- B. Removes system-specific details, Linux only
- C. Backs up template, all OS
- D. Compresses template, all OS

45. When creating a VM from a template, what do you NOT need to provide?

- A. VM name
- B. Instance images (disks are already from template)
- C. Network interface
- D. Memory size

EXPORTING VMs

46. What file format is used when exporting a VM?

- A. VMDK
- B. OVA
- C. VHD
- D. ISO

47. Where must you create a directory for VM export?

- A. On the engine host

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- B. On the KVM host where export will happen
 - C. On the storage domain
 - D. On the client machine

48. What permissions should be set on the export directory?

- A. 600 (owner only)
- B. 755 (owner full, others read/execute)
- C. 777 (full permissions for all)
- D. 644 (owner read/write, others read)

49. How do you initiate VM export in the Administration Portal?

- A. File -> Export
- B. Select VM -> Options menu (three dots) -> Export as OVA
- C. Right-click -> Export
- D. VM -> Tools -> Export

50. Can you export a running VM?

- A. No, it must be shut down first
- B. Yes, VMs can be exported while running
- C. Only if memory is saved
- D. Only test VMs

REVERTING TO SNAPSHOTS

51. To revert a VM to a snapshot, what must be the VM state?

- A. Running
- B. Shut down
- C. Paused
- D. Any state

52. What two actions are required to revert to a snapshot? (Choose 2)

- A. Select snapshot -> Preview
- B. Delete the VM
- C. Click Commit button
- D. Restart engine

53. After reverting to a snapshot, what happens to changes made after the snapshot was taken?

- A. They are preserved
- B. They are lost
- C. They are archived
- D. They are encrypted

54. In the practice, what file was created post-snapshot to demonstrate reversion?

-
- A. test-file
 - B. post-snap
 - C. snapshot-test
 - D. demo-file

55. After reverting to the snapshot, was the post-snap file still present?

- A. Yes, it remained
- B. No, it was gone because VM reverted to snapshot state
- C. It was moved to backup
- D. It was encrypted

Answer Key — Virtual Machine Administration

- 1. A, C - Pre-allocated (thick), Sparse (thin)
- 2. B - Thick provisioning
- 3. B - Physical space allocated only as data is written
- 4. A, C - QCOW2, Raw
- 5. B - QEMU Copy On Write version 2
- 6. C - QCOW2
- 7. B - A file with initial size close to 0
- 8. B - A block device with initial size smaller than defined size
- 9. B - Yes, they can be sparse files or pre-allocated
- 10. C - Only pre-allocated
- 11. B - Replaces user data with zeros when deleted
- 12. B - Does nothing because file system ensures no data exists
- 13. C - If the virtual disk contained sensitive data
- 14. B - False
- 15. B - engine-config --set SANWipeAfterDelete=true
- 16. B - Restart the ovirt-engine service
- 17. B - Performance degradation and prolonged delete times
- 18. B - Virtual disks accessed simultaneously by multiple VMs
- 19. B - Only cluster-aware VMs
- 20. B - Data corruption
- 21. B - Clustered database servers or highly available services
- 22. B - Compute -> Virtual Machines
- 23. B - Blank
- 24. A, C - Pre-allocated, Thin provisioning (sparse)
- 25. C - VirtIO-SCSI
- 26. B - CD-ROM
- 27. B - Hard disk
- 28. C - Escape

-
29. C - Shift+F12
 30. B - Point-in-time capture of VM state
 31. B - No, should not be used as primary backup
 32. B - Stored on same storage; if storage fails, snapshots inaccessible
 33. B - Before testing updates, configuration changes, patches
 34. B - Delete them promptly
 35. B - Increased storage usage and performance degradation
 36. B - Latest guest agent package (qemu-guest-agent)
 37. B - Compute -> Virtual Machines -> select VM -> Create Snapshot
 38. B - Memory state of the VM
 39. B - VM will be paused while saving memory
 40. B - Pre-configured VM for quick deployment
 41. B - Select snapshot -> "Make Template" button
 42. B - Compute -> Templates
 43. B - Allow all users to access this template
 44. B - Removes system-specific details, Linux only
 45. B - Instance images (disks already from template)
 46. B - OVA
 47. B - On the KVM host where export will happen
 48. C - 777 (full permissions for all)
 49. B - Select VM -> Options menu -> Export as OVA
 50. B - Yes, VMs can be exported while running
 51. B - Shut down
 52. A, C - Preview, then Commit
 53. B - They are lost
 54. B - post-snap
 55. B - No, it was gone

Score Interpretation

- 51-55 correct: Excellent - Comprehensive mastery
- 46-50 correct: Very Good - Strong understanding
- 41-45 correct: Good - Ready with review
- 36-40 correct: Fair - Review weak areas
- Below 36: Needs comprehensive review

Key Exam Topics

- Two storage types: Pre-allocated (thick) vs Sparse (thin)
- Two disk formats: QCOW2 (features) vs Raw (simple)
- QCOW2 on NFS: file; on SAN: block device
- Raw on NFS: sparse or pre-allocated; on SAN: pre-allocated only
- Wipe after delete: replaces data with zeros, default=false
- Shareable disks: only for cluster-aware VMs, risk of corruption

-
- VM creation: Blank template for new installs
 - Boot devices: CD-ROM first (install), Hard disk after
 - Snapshots: NOT primary backup, stored on same storage
 - Delete snapshots promptly after use
 - Guest agent required before snapshots
 - Templates: created from snapshots
 - Export format: OVA file
 - Revert: VM must be shut down, Preview then Commit

Section 10: Events & Logs Management

45 Questions

LOG FILES OVERVIEW

1. What are the three main categories of log files in OLVM? (Choose 3)

- A. Engine installation log files
- B. Network log files
- C. oVirt engine log files
- D. Host log files
- E. User activity log files

2. Where are engine installation log files located?

- A. /var/log/ovirt
- B. /var/log/ovirt-engine
- C. /var/log/engine
- D. /etc/ovirt-engine/logs

3. Which two key log files are in the engine installation logs? (Choose 2)

- A. engine-setup.log
- B. engine-install.log
- C. engine-setup-summary.log
- D. installation.log

4. What does the engine-setup.log file contain?

- A. Summary only
- B. Detailed information about each step and action during setup or upgrade
- C. Error messages only
- D. Network configurations

5. Where are oVirt engine log files located?

- A. /var/log/engine
- B. /var/log/ovirt-engine
- C. /var/log/olvm
- D. /etc/ovirt/logs

ENGINE LOG FILES

6. Which four key log files are part of oVirt engine logs? (Choose 4)

- A. engine.log
- B. server.log
- C. audit.log
- D. boot.log
- E. network.log
- F. storage.log

7. What does the engine.log file record?

- A. User authentication only
- B. General operation information about oVirt engine (starting/stopping services, API requests)
- C. Boot process only
- D. Audit trails only

8. What does the server.log file contain?

- A. Audit information
- B. Detailed information about server-side processes
- C. Boot messages only
- D. User activities

9. What does the audit.log file contain?

- A. General operations
- B. Security events and audit trails (user actions, authentication, permission changes)
- C. Boot process
- D. Network events

10. What does the boot.log file record?

- A. User logins
- B. The booting process of oVirt engine
- C. Shutdown events
- D. API calls

HOST LOG FILES

11. Where is the libvirt log file located?

- A. /var/log/libvirt
- B. /var/log/messages
- C. /var/log/vdsm
- D. /var/log/hosts

12. What does the SPM lock.log file detail?

- A. User logins

-
- B. Host ability to obtain a lease on storage pool management
 - C. Network traffic
 - D. VM migrations

13. Where is the SPM lock.log file located?

- A. /var/log/spm
- B. /var/log/vdsm/spm-lock.log
- C. /var/log/ovirt
- D. /var/log/storage

14. Where is the VDSM service log file located?

- A. /var/log/vdsm/vdsm.log
- B. /var/log/vdsm.log
- C. /var/log/messages
- D. /var/log/ovirt/vdsm

15. Where is the host deployment log file located?

- A. /var/log/deploy
- B. /tmp/ovirt-host-deploy-<date>
- C. /var/log/ovirt-deploy
- D. /var/log/vdsm/deploy

16. Where are VM import log files located?

- A. /var/log/imports
- B. /var/log/vdsm/import/import-<UUID>-<date>
- C. /var/log/vm-import
- D. /tmp/imports

17. What does the supervdsm.log file record?

- A. All VDSM activities
- B. VDSM tasks executed with superuser permissions
- C. VM migrations only
- D. Network configurations

18. Where is the supervdsm.log file located?

- A. /var/log/supervdsm
- B. /var/log/vdsm/supervdsm.log
- C. /var/log/root
- D. /var/log/admin

19. What does the mom.log file contain?

- A. Network bandwidth
- B. Memory overcommitment information and MoM (Memory Overcommitment Manager) activities

-
- C. User activities
 - D. Storage usage

20. Where is the upgrade.log file located?

- A. /var/log/upgrade
- B. /var/log/vdsm/upgrade.log
- C. /var/log/ovirt/upgrade
- D. /tmp/upgrade

EVENT NOTIFICATION

21. What notification methods does OLVM support? (Choose 2)

- A. Email notifications
- B. SMS notifications
- C. SNMP traps
- D. Slack integration

22. Where is the event notifier configuration file located?

- A. /etc/ovirt-engine/notifier.conf
- B. /etc/ovirt-engine-notifier/ovirt-engine-notifier.conf
- C. /etc/ovirt-engine/services/ovirt-engine-notifier/ovirt-engine-notifier.conf
- D. /var/lib/ovirt/notifier.conf

23. What parameter specifies the SMTP mail server address?

- A. smtp_server
- B. mail_server
- C. email_server
- D. notification_server

24. Which three port numbers are commonly used for mail servers? (Choose 3)

- A. 25 (plain SMTP)
- B. 465 (SMTP with SSL)
- C. 587 (SMTP with TLS)
- D. 8080 (HTTP)
- E. 3306 (MySQL)

25. What parameter specifies the username for mail server authentication?

- A. smtp_user
- B. mail_user
- C. email_user
- D. auth_user

26. What parameter contains the password for mail authentication?

-
- A. smtp_password
 - B. mail_password
 - C. email_password
 - D. auth_password

27. What parameter specifies the engine's process ID?

- A. engine_id
- B. engine_pid
- C. process_id
- D. ovirt_pid

SNMP TRAP CONFIGURATION

28. Which two SNMP versions does oVirt support? (Choose 2)

- A. SNMP version 1
- B. SNMP version 2
- C. SNMP version 3
- D. SNMP version 4

29. What are the three SNMP security levels? (Choose 3)

- A. NoAuthNoPriv
- B. BasicAuth
- C. AuthNoPriv
- D. AuthPriv
- E. FullAuth

30. What does NoAuthNoPriv security level provide?

- A. Full encryption
- B. SNMP traps sent without authentication or encryption (clear text)
- C. Password authentication only
- D. Encryption only

31. What does AuthNoPriv security level provide?

- A. No security
- B. Password authentication but no encryption (clear text transmission)
- C. Full encryption
- D. Authentication and encryption

32. What does AuthPriv security level provide?

- A. No security
- B. Password authentication only
- C. Password authentication and encryption (highest security)
- D. Encryption only

33. What information is required to configure SNMP managers? (Choose 3)

- A. IP address or FQDN of SNMP manager
- B. MAC address
- C. SNMP version supported
- D. Community string or SNMP v3 credentials
- E. Operating system version

LOG COLLECTOR TOOL

34. What is the oVirt log collector diagnostic tool used for?

- A. Creating new logs
- B. Collecting comprehensive logs from manager and attached hosts for troubleshooting
- C. Deleting old logs
- D. Encrypting logs

35. What package should be installed on manager and hosts for system diagnostics?

- A. syslog
- B. sos (sosreport)
- C. logrotate
- D. rsyslog

36. What command is used to install the ovirt-log-collector package?

- A. dnf install log-collector
- B. yum install ovirt-log-collector
- C. dnf install olvm-logs
- D. yum install log-diagnostic

37. What does the ovirt-log-collector tool collect? (Choose 3)

- A. SOS report from engine
- B. Engine log files
- C. Browser history
- D. Engine PostgreSQL database logs
- E. User passwords

38. What parameter skips PostgreSQL database logs when collecting?

- A. --skip-database
- B. --no-postgresql
- C. --exclude-db
- D. --no-database

39. What parameter skips host log collection?

- A. --skip-hosts

-
- B. --no-hosts
 - C. --no-hypervisors
 - D. --exclude-hosts

40. What parameter collects logs per hypervisor per cluster?

- A. --cluster-mode
- B. --per-cluster
- C. --hypervisors-per-cluster
- D. --cluster-hypervisors

POSTGRESQL DATABASE

41. What are the two PostgreSQL databases in OLVM? (Choose 2)

- A. engine
- B. admin
- C. ovirt_engine_history
- D. manager
- E. olvm_data

42. What does the engine database contain?

- A. Historical data only
- B. Active operational data
- C. Backup data
- D. Archived logs

43. What does the ovirt_engine_history database contain?

- A. Active data only
- B. Historical configuration information and statistical metrics collected over time
- C. User accounts
- D. Network configurations

44. What command switches to the postgres user?

- A. su postgres
- B. su - postgres
- C. sudo postgres
- D. switch postgres

45. What command connects to the engine database?

- A. psql engine
- B. psql -d engine
- C. postgres -d engine
- D. connect engine

46. What PostgreSQL command lists all databases?

- A. show databases
- B. list databases
- C. \l
- D. \databases

47. What PostgreSQL command shows connection information?

- A. \info
- B. \connection
- C. \conninfo
- D. \conn

48. What PostgreSQL command lists all tables?

- A. show tables
- B. \tables
- C. \dt
- D. list tables

49. What PostgreSQL command describes a specific table structure?

- A. describe tablename
- B. \d tablename
- C. show tablename
- D. \desc tablename

50. What PostgreSQL command quits the database prompt?

- A. exit
- B. quit
- C. \q
- D. \quit

GRAFANA MONITORING

51. What command shows the Grafana server version?

- A. grafana --version
- B. grafana-server -v
- C. grafana version
- D. grafana-server --version

52. What port does Grafana typically use?

- A. 8080
- B. 3000
- C. 5432

D. 9090

53. What configuration must be enabled to avoid "origin not allowed" errors in Grafana?

- A. cors_enabled
- B. proxy_preserve_host
- C. allow_origin
- D. enable_proxy

54. What database does Grafana connect to for OLVM monitoring?

- A. engine database
- B. ovirt_engine_history (DWH database)
- C. admin database
- D. grafana database

55. What permission level should be granted to the Grafana database user?

- A. Full admin access
- B. Read-only (SELECT) permissions on DWH database
- C. Write access
- D. Root access

Answer Key — Events & Logs Management

1. A, C, D - Engine installation, oVirt engine, Host log files
2. B - /var/log/ovirt-engine
3. A, C - engine-setup.log, engine-setup-summary.log
4. B - Detailed information about each step during setup/upgrade
5. B - /var/log/ovirt-engine
6. A, B, C, D - engine.log, server.log, audit.log, boot.log
7. B - General operation information (services, API requests)
8. B - Detailed information about server-side processes
9. B - Security events and audit trails
10. B - The booting process of oVirt engine
11. B - /var/log/messages
12. B - Host ability to obtain lease on storage pool management
13. B - /var/log/vdsm/spm-lock.log
14. A - /var/log/vdsm/vdsm.log
15. B - /tmp/ovirt-host-deploy-<date>
16. B - /var/log/vdsm/import/import-<UUID>-<date>
17. B - VDSM tasks with superuser permissions
18. B - /var/log/vdsm/supervdsm.log
19. B - Memory overcommitment and MoM activities

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20. B - /var/log/vdsm/upgrade.log
 21. A, C - Email notifications, SNMP traps
 22. C - /etc/ovirt-engine/services/ovirt-engine-notifier/ovirt-engine-notifier.conf
 23. B - mail_server
 24. A, B, C - 25, 465, 587
 25. B - mail_user
 26. B - mail_password
 27. B - engine_pid
 28. B, C - SNMP version 2, version 3
 29. A, C, D - NoAuthNoPriv, AuthNoPriv, AuthPriv
 30. B - No authentication or encryption (clear text)
 31. B - Password authentication but no encryption
 32. C - Password authentication and encryption
 33. A, C, D - IP/FQDN, SNMP version, Credentials
 34. B - Collecting logs from manager and hosts for troubleshooting
 35. B - sos (sosreport)
 36. B - yum install ovirt-log-collector
 37. A, B, D - SOS report, Engine logs, PostgreSQL logs
 38. B - --no-postgresql
 39. C - --no-hypervisors
 40. C - --hypervisors-per-cluster
 41. A, C - engine, ovirt_engine_history
 42. B - Active operational data
 43. B - Historical configuration and statistical metrics
 44. B - su - postgres
 45. B - psql -d engine
 46. C - \l
 47. C - \conninfo
 48. C - \dt
 49. B - \d tablename
 50. C - \q
 51. B - grafana-server -v
 52. B - 3000
 53. B - proxy_preserve_host
 54. B - ovirt_engine_history (DWH database)
 55. B - Read-only (SELECT) permissions

Score Interpretation

- 51-55 correct: Excellent - Comprehensive mastery
- 46-50 correct: Very Good - Strong understanding
- 41-45 correct: Good - Ready with review
- 36-40 correct: Fair - Review weak areas

Below 36: Needs comprehensive review

Key Exam Topics

- Three log categories: Engine install, oVirt engine, Host logs
- Engine logs location: /var/log/ovirt-engine
- Four key engine logs: engine.log, server.log, audit.log, boot.log
- Host logs: messages, vdsd.log, spm-lock.log, supervdsm.log, mom.log
- Event notification: email and SNMP traps
- Mail parameters: mail_server, mail_port, mail_user, mail_password
- SNMP versions: 2 and 3
- SNMP security: NoAuthNoPriv, AuthNoPriv, AuthPriv
- Log collector: yum install ovirt-log-collector
- Log collector flags: --no-postgresql, --no-hypervisors
- Two databases: engine (active), ovirt_engine_history (historical)
- PostgreSQL commands: \l (list), \dt (tables), \d (describe), \q (quit)
- psql -d engine (connect to database)
- Grafana: DWH database, port 3000, proxy_preserve_host

Section 11: Recovery Management

40 Questions

BACKUP FUNDAMENTALS

1. What tool is used to backup and restore OLVM?

- A. olvm-backup
- B. engine-backup
- C. ovirt-backup
- D. manager-backup

2. What does the engine-backup tool backup? (Choose 2)

- A. Engine database
- B. Virtual machine disk images
- C. Configuration files
- D. User home directories

3. Can the engine-backup tool be used while the engine is active?

- A. No, engine must be stopped
- B. Yes, backups can be performed while engine is active
- C. Only in maintenance mode
- D. Only during off-peak hours

4. What user must you be logged in as to run engine-backup?

- A. admin
- B. postgres
- C. root
- D. ovirt

5. What parameter specifies whether to perform backup or restore?

- A. --action
- B. --mode
- C. --operation
- D. --task

BACKUP SCOPE OPTIONS

6. What is the default scope when running engine-backup without parameters?

-
- A. files
 - B. db
 - C. all
 - D. config

7. What does scope=all backup?

- A. Database only
- B. Files only
- C. Full backup of all databases and configuration files
- D. Configuration files only

8. What does scope=files backup?

- A. All files and databases
- B. Only the files on the system
- C. Only database files
- D. Only log files

9. What does scope=db backup?

- A. All databases
- B. Only the engine database (active database)
- C. Only data warehouse database
- D. Configuration files

10. What does scope=dwhdb backup?

- A. Engine database
- B. Only the data warehouse database (historical data)
- C. All databases
- D. Configuration files

11. Which other database scopes are available? (Choose 2)

- A. cinderlib_db
- B. user_db
- C. grafana_db
- D. admin_db

BACKUP PARAMETERS

12. What parameter specifies the path and name of the backup file?

- A. --backup
- B. --file
- C. --output
- D. --archive

13. What parameter specifies where to write backup/restore operation logs?

- A. --logfile
- B. --log
- C. --output-log
- D. --recording

14. What is the default file location for engine-backup?

- A. /root
- B. /tmp
- C. /var/lib/ovirt-engine-backup
- D. /home/admin

15. What is the default log file location?

- A. /var/log/ovirt-engine
- B. /tmp/engine-backup.log
- C. /var/log/engine-backup.log
- D. /root/backup.log

RESTORE OPERATIONS

16. To restore a backup, what mode must be specified?

- A. --mode=recover
- B. --mode=restore
- C. --mode=import
- D. --mode=recovery

17. Before restoring a backup, what should be done to the engine service?

- A. Nothing required
- B. Stop the engine service
- C. Restart the engine service
- D. Enable debug mode

18. After restoring a backup, what command must be run to reconfigure the engine?

- A. engine-config
- B. engine-setup
- C. ovirt-setup
- D. engine-restore

19. If database credentials are not known during restore, what must you do?

- A. Reinstall the engine
- B. Change the password for the engine database owner using PostgreSQL
- C. Use default credentials

D. Skip database restore

20. What PostgreSQL command changes the database owner password?

- A. UPDATE USER SET PASSWORD
- B. ALTER USER username ENCRYPTED PASSWORD 'newpassword'
- C. CHANGE PASSWORD FOR username
- D. SET PASSWORD username = 'newpassword'

DATA WAREHOUSE MIGRATION

21. What are two methods to migrate the data warehouse? (Choose 2)

- A. Migrate DWH service only, keep database on engine
- B. Delete and recreate
- C. Migrate both DWH database and service to new machine
- D. Export and import

22. What Oracle Linux version must the new DWH server have?

- A. Oracle Linux 7
- B. Oracle Linux 8
- C. Oracle Linux 9
- D. Any version

23. Which modules must be enabled on the new DWH server? (Choose 4)

- A. javapackages-tools
- B. postgresql:13 or postgresql:12
- C. mod_auth_openidc:2.3
- D. nodejs:14
- E. python:3.8

24. What command enables a module in RHEL/Oracle Linux 8?

- A. yum module enable
- B. dnf module enable
- C. module enable
- D. enable-module

25. What command synchronizes packages to latest versions?

- A. dnf update
- B. dnf upgrade
- C. dnf distro-sync
- D. dnf sync

26. What must be backed up before migrating DWH?

- A. Only the database

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- B. DWH database and configuration files
 - C. Only configuration files
 - D. Virtual machine data

27. What package must be installed on the new DWH machine?

- A. ovirt-engine
- B. ovirt-engine-tools-backup
- C. postgresql-backup
- D. dwh-tools

28. Which PostgreSQL packages must be installed? (Choose 2)

- A. postgresql-server
- B. postgresql-client
- C. postgresql-contrib
- D. postgresql-backup

29. After installing PostgreSQL, what must be done before restoring?

- A. Nothing
- B. Enable and start PostgreSQL service
- C. Configure firewall only
- D. Create database manually

30. What scope is used to restore DWH to the new machine?

- A. --scope=all
- B. --scope=dwhdb
- C. --scope=files,grafana_db,dwhdb
- D. --scope=database

RENAMING THE ENGINE

31. What tool is used to rename the OLVM engine?

- A. engine-rename
- B. ovirt-rename
- C. ovirt-engine-rename
- D. manager-rename

32. Where is the ovirt-engine-rename command located?

- A. /usr/bin
- B. /usr/sbin
- C. /usr/share/ovirt-engine/setup/bin
- D. /etc/ovirt-engine

33. What must be updated before renaming the engine? (Choose 2)

-
- A. DNS A records
 - B. MAC addresses
 - C. DNS PTR records
 - D. IP addresses

34. If DHCP is used, what must be updated?

- A. Nothing
- B. DHCP server configuration to reflect new hostname
- C. Static IP addresses
- D. DNS only

35. What must be updated on the engine server itself?

- A. IP address
- B. Hostname to new FQDN
- C. Network interface
- D. Storage domain

DISASTER RECOVERY - ACTIVE-ACTIVE

36. What configuration is used for active-active disaster recovery?

- A. Two separate environments
- B. Stretched cluster configuration
- C. Master-slave replication
- D. Backup-only site

37. In active-active DR, are both sites operational?

- A. No, one is standby
- B. Yes, both sites maintain active operations simultaneously
- C. Only during failover
- D. Depends on load

38. What type of network is required for stretched clusters?

- A. Layer 3 only
- B. Layer 2 network with sufficient bandwidth and low latency
- C. VPN only
- D. Satellite links

39. What storage type is typically required for stretched clusters?

- A. Local storage only
- B. Shared storage (SAN or NAS) accessible from both sites
- C. Cloud storage
- D. Tape storage

40. What happens if one site fails in active-active DR?

- A. All VMs stop
- B. Cluster migrates all VMs to the other site
- C. Manual intervention required
- D. Data is lost

DISASTER RECOVERY - ACTIVE-PASSIVE

41. How is active-passive DR implemented?

- A. Stretched cluster
- B. Two separate OLVM environments (primary and secondary sites)
- C. Single site with backups
- D. Cloud-only deployment

42. In active-passive DR, which site runs production workloads?

- A. Both sites
- B. Primary site
- C. Secondary site
- D. Alternates between sites

43. What is the role of the secondary site in active-passive DR?

- A. Active workload processing
- B. Standby/backup location that replicates data from primary
- C. Development only
- D. Storage only

44. What is a benefit of active-passive DR?

- A. Higher performance
- B. Cost-effective, utilizing standby resources only when needed
- C. Faster than active-active
- D. No replication needed

45. What must be replicated from primary to secondary site?

- A. Only databases
- B. Critical data, OLVM configurations, and VM images
- C. Only configuration files
- D. User accounts only

ENGINE CLEANUP & RECOVERY

46. What command uninstalls the OLVM engine?

- A. engine-remove

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- B. engine-cleanup
 - C. engine-uninstall
 - D. ovirt-remove

47. What happens when you run engine-cleanup?

- A. Only stops the service
- B. Stops engine service and removes all installed ovirt components
- C. Backs up data
- D. Restarts the engine

48. After running engine-cleanup, what happens to the Administration Portal?

- A. Still accessible
- B. Site cannot be reached
- C. Shows error page
- D. Redirects to new URL

49. What parameter restores file permissions during restore?

- A. --restore-perms
- B. --restore-permissions
- C. --fix-permissions
- D. --permissions

50. After restoring from backup, what appears in the Administration Portal?

- A. Empty environment
- B. All previous data, VMS, and configurations are restored
- C. Only database is restored
- D. Only configuration files

Answer Key — Recovery Management

1. B - engine-backup
2. A, C - Engine database, Configuration files
3. B - Yes, backups can be performed while engine is active
4. C - root
5. B - --mode
6. C - all
7. C - Full backup of all databases and configuration files
8. B - Only the files on the system
9. B - Only the engine database
10. B - Only the data warehouse database
11. A, C - cinderlib_db, grafana_db

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12. B - --file
 13. B - --log
 14. C - /var/lib/ovirt-engine-backup
 15. C - /var/log/engine-backup.log
 16. B - --mode=restore
 17. B - Stop the engine service
 18. B - engine-setup
 19. B - Change password for engine database owner using PostgreSQL
 20. B - ALTER USER username ENCRYPTED PASSWORD 'newpassword'
 21. A, C - Migrate service only, or Migrate both database and service
 22. B - Oracle Linux 8
 23. A, B, C, D - javapackages-tools, postgresql, mod_auth_openidc, nodejs
 24. B - dnf module enable
 25. C - dnf distro-sync
 26. B - DWH database and configuration files
 27. B - ovirt-engine-tools-backup
 28. A, C - postgresql-server, postgresql-contrib
 29. B - Enable and start PostgreSQL service
 30. C - --scope=files,grafana_db,dwhdb
 31. C - ovirt-engine-rename
 32. C - /usr/share/ovirt-engine/setup/bin
 33. A, C - DNS A records, DNS PTR records
 34. B - DHCP server configuration to reflect new hostname
 35. B - Hostname to new FQDN
 36. B - Stretched cluster configuration
 37. B - Yes, both sites maintain active operations simultaneously
 38. B - Layer 2 network with sufficient bandwidth and low latency
 39. B - Shared storage (SAN or NAS) accessible from both sites
 40. B - Cluster migrates all VMs to the other site
 41. B - Two separate OLVM environments
 42. B - Primary site
 43. B - Standby/backup location that replicates data
 44. B - Cost-effective, resources only when needed
 45. B - Critical data, OLVM configurations, VM images
 46. B - engine-cleanup
 47. B - Stops engine and removes all installed ovirt components
 48. B - Site cannot be reached
 49. B - --restore-permissions
 50. B - All previous data, VMs, configurations are restored

Score Interpretation

46-50 correct: Excellent - Comprehensive mastery

41-45 correct: Very Good - Strong understanding

36-40 correct: Good - Ready with review

31-35 correct: Fair - Review weak areas

Below 31: Needs comprehensive review

Key Exam Topics

- engine-backup: backup and restore tool
- Must run as root user
- Default scope: all (full backup)
- Scope options: all, files, db, dwhdb, grafana_db, cinderlib_db
- Key parameters: --mode, --file, --log
- Restore: --mode=restore, then run engine-setup
- DWH migration: Oracle Linux 8, enable modules (postgresql, nodejs, etc.)
- DWH restore scope: files, grafana_db, dwhdb
- Renaming: ovirt-engine-rename, update DNS A/PTR records
- Active-active DR: stretched cluster, both sites active, Layer 2 network
- Active-passive DR: two separate environments, primary active
- engine-cleanup: removes all components
- Restore includes: --restore-permissions parameter

End of Practice Test Collection

515 questions across 11 exam objectives

Good luck on Exam 1Z0-1170!